

Exploring Triaxiality in the Lead Region



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Abstract

Nuclear deformation is thought to be driven by interactions between valence nucleons, however the underlying mechanism of this process is not yet fully understood. This project investigates the influence of deformation on the spectroscopic properties of three nuclei in the lead region. A Particle-plus-Triaxial-Rotor Model (PTRM) with a modified harmonic oscillator potential is used to predict several nuclear properties, including the ground state magnetic dipole moment and the spins and energies of the excited states, when assuming a range of quadrupole ($\varepsilon=[0.0, 0.5]$) and triaxial ($\gamma = [0, 60]^\circ$) shapes. The results are compared with experimental values to deduce the ground state deformations of these nuclei. A near-spherical shape of $^{207}_{82}\text{Pb}_{125}$ is shown to be well described by the PTRM with single particle excitations. Meanwhile, highly triaxial deformations of $\varepsilon = [0.12 \pm 0.06], \gamma = [35 \pm 5]^\circ$ and $\varepsilon = [0.28 \pm 0.04], \gamma = [27 \pm 3]^\circ$ are proposed as good descriptions of $^{179}_{79}\text{Au}_{100}$ and $^{177}_{78}\text{Pt}_{99}$ respectively in this model. A Python interface to the PTRM codes is developed, improving workflow efficiency and ease of use for future users of the model, and facilitating further work in this direction.

1 Introduction

The isotope $^{208}_{82}\text{Pb}_{126}$ is of special interest to nuclear physics, due to its doubly-magic number of protons and neutrons. In the nuclear shell model, nucleons fill quantised energy levels similarly to electrons in atomic shells. These energy levels are not evenly spaced, as seen in Fig. 1. Certain “magic” numbers of nucleons fill the energy levels exactly up to large energy gaps, known as shell closures. These magic numbers occur at proton¹ and neutron numbers $Z, N = 2, 8, 20, 28, 50, 82, 126$. Another nucleon added on top of a shell closure will be weakly bound, making these closed-shell nuclei relatively more stable. In the case of $^{208}_{82}\text{Pb}_{126}$, both its proton and neutron numbers are magic (see Fig. 1(a)), and this is the heaviest known doubly-magic nucleus.

The closed-shell nature of this isotope gives it a spherical shape at low energies, typical of magic nuclei. This is caused by the absence of valence nucleons in the ground state, where the large energy required for any single-particle excitations across the shell gap prevents the creation of valence nucleons at low energy. Without residual interactions between valence nucleons, there are no strong proton-neutron correlations to drive deformation [1]. This is illustrated in Fig. 1(a), where a sharp potential minimum maintains the spherical shape of the nucleus.

The stability of this nucleus makes it easier to study experimentally, and the simplicity of its closed-shell structure makes it a useful benchmark for testing many theoretical models [2]. It has therefore been thoroughly studied. Its neighbouring isotopes $^{209}_{82}\text{Pb}_{127}$ and $^{207}_{82}\text{Pb}_{125}$ are also thought to be near-spherical [3], being effectively a single particle or hole coupled to the doubly-magic core. This picture can carry slightly further away from the closed-shell, by coupling as many as three particles or holes [4], however the further one looks, the more likely a nucleus is to be deformed.

For example, the isotope $^{186}_{82}\text{Pb}_{104}$ is highly neutron deficient, with $N = 104$ lying at the neutron mid-shell. The large number of valence neutrons allows for strong proton-neutron interactions when protons are excited across the shell gap, resulting in the formation of low-lying deformed excited states [5, 6]. Fig. 1(b) illustrates the presence of valence neutrons in the energy levels of this nucleus, and how this can create local minima in the potential energy surface at non-spherical deformations.

The exact mechanism by which residual interactions between valence nucleons induce deformation is not yet fully understood [7]. By studying how nucleon numbers impact on the shape adopted by a nucleus, we can gain insight into the role of proton-neutron interactions. In order to motivate and test theoretical models, we need extensive data on the different shapes adopted across the nuclear chart, their deformation strengths, and the rate of variation as N and Z change.

The axially-symmetric quadrupole deformation is common across the nuclear landscape [1]. It is quantified by the deformation parameter (β_2) given by:

$$\beta_2 = \frac{4}{3} \sqrt{\frac{\pi}{5}} \left(\frac{c - a}{R_{av}} \right) \quad (1)$$

where c is the length of the nucleus along the symmetry axis, a is its length along the perpendicular axes, and R_{av} is the average radius (see Fig. 2). A spherical shape is

¹Note that there are no known elements with a proton number as high as $Z = 126$, therefore the proton shell closures have only been confirmed up to $Z = 82$.

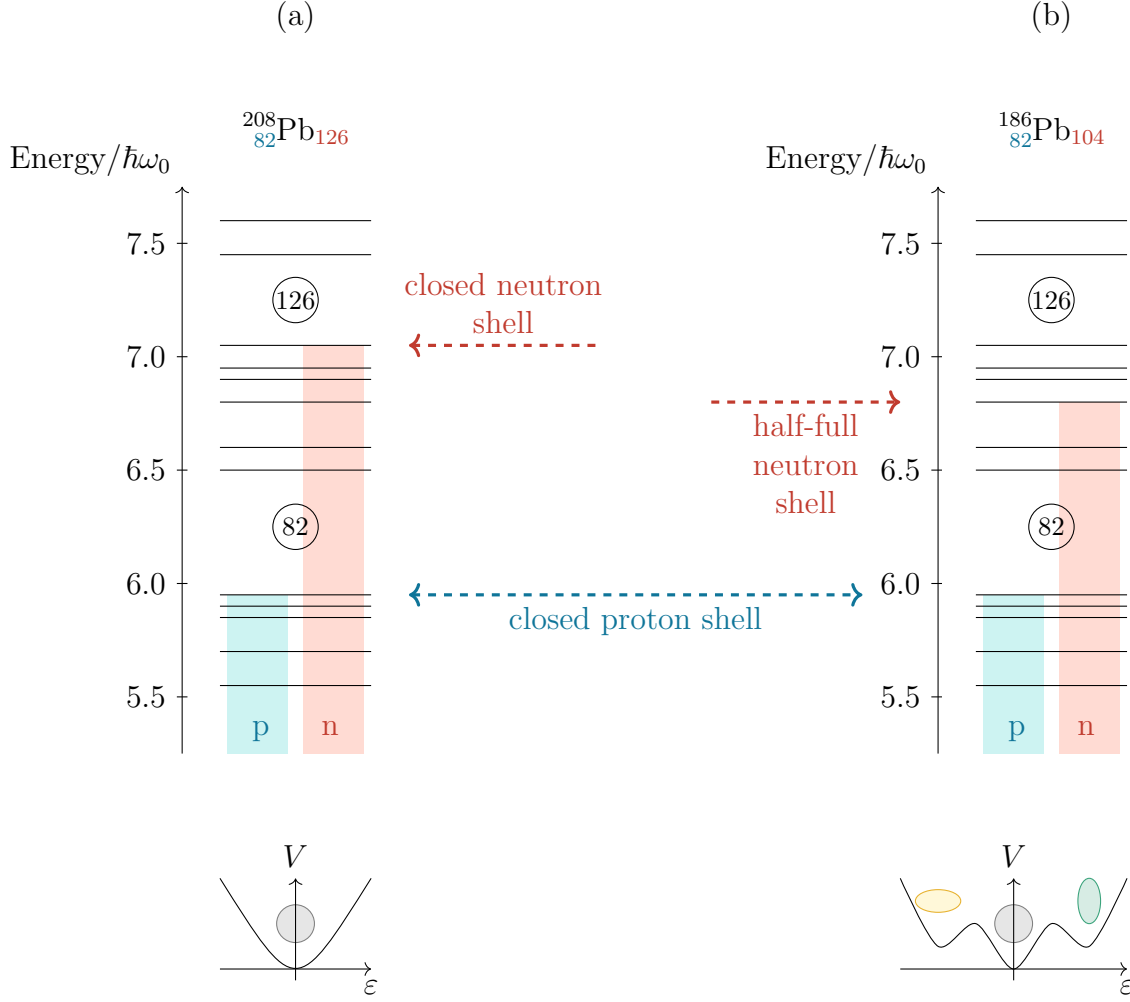


Figure 1: An illustration of how the different numbers of valence neutrons in (a) $^{208}_{82}\text{Pb}_{126}$ and (b) $^{186}_{82}\text{Pb}_{104}$ impact their shapes in the ground state and excited states. The shaded regions show how the nucleons fill the energy levels (blue for protons and red for neutrons). Magic numbers are indicated in circles. The schematic potential energy graphs suggest how deformed excited states may be more easily accessed in (b).

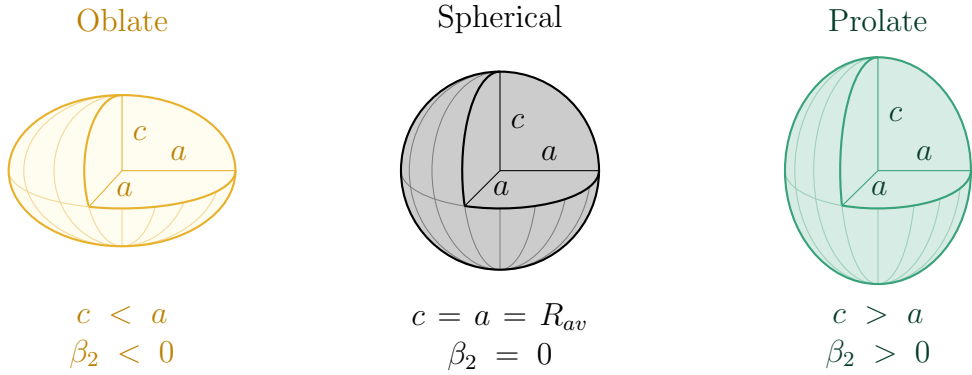


Figure 2: The quadrupole deformation can be classified as oblate (symmetry axis compressed) or prolate (symmetry axis elongated). The oblate and prolate shapes illustrated here have $|\beta_2| \approx 0.2$.

described by $\beta_2 = 0$. Negative β_2 describes an oblate shape, where the symmetry axis is compressed, while positive β_2 describes a prolate shape, where the symmetry axis is elongated.

Certain regions of the nuclear chart exhibit an exotic form of deformation known as *triaxiality*. This non-axially-symmetric deformation describes the case where all three principle axes of the nucleus have different lengths (see Fig. 3). If we define the deformation in terms of spherical harmonics, $Y_{lm}(\theta, \phi)$, then the quadrupole deformation ($l = 2$) is given by:

$$R = R_{av} \left(1 + \sum_{m=-2}^2 \alpha_{2m} Y_{2m}(\theta, \phi) \right) \quad (2a)$$

with coefficients:

$$\alpha_{2,0} = \beta_2 \cos(\gamma) \quad (2b)$$

$$\alpha_{2,\pm 1} = 0 \quad (2c)$$

$$\alpha_{2,\pm 2} = \frac{1}{\sqrt{2}} \beta_2 \sin(\gamma) \quad (2d)$$

Here we have introduced the triaxiality parameter, γ , which describes the deviation from axial symmetry. An alternative convention, first described by Nilsson [8], defines the quadrupole deformation as ε , which is related to β_2 by:

$$\beta_2 \simeq \sqrt{\frac{16\pi}{5}} \left(\frac{\varepsilon}{3} + \frac{\varepsilon^2}{9} + \frac{\varepsilon^3}{27} + \dots \right) \quad (3)$$

The parameters ε and γ can then be used to define the oscillator frequencies ($\omega_x, \omega_y, \omega_z$) of the 3D harmonic potential, where m is the nucleon mass, and (x, y, z) are the principal axes of the nucleus.

$$V = \frac{1}{2} m \left[\omega_x^2 x^2 + \omega_y^2 y^2 + \omega_z^2 z^2 \right] \quad (4a)$$

$$\omega_k = \omega_0(\varepsilon, \gamma) \left[1 - \frac{2}{3} \varepsilon \cos(\gamma + q) \right] \begin{cases} q = \frac{2\pi}{3} & \text{for } k = x \\ q = -\frac{2\pi}{3} & \text{for } k = y \\ q = 0 & \text{for } k = z \end{cases} \quad (4b)$$

It should be noted that ω_0 is a function of ε and γ , as determined by the volume conservation condition [9]:

$$\omega_x \omega_y \omega_z = \left[\overset{o}{\omega}_0 \right]^3 \equiv \left[\omega_0(\varepsilon = 0, \gamma = 0) \right]^3 \quad (4c)$$

This ensures that the total volume enclosed by a nuclear potential surface is independent of deformation, and determined only by the number of nucleons, due to the short range of the nuclear force.

These deformation parameters are not uniquely valued; a single sector in the range $\varepsilon \geq 0$ and $\gamma = [0, 60]^\circ$ is sufficient to describe all possible deformations (see Fig. 3). The remaining sectors of γ simply correspond to a different choice of the principle axes (x, y, z) . By fixing $\varepsilon \geq 0$, we distinguish between “prolate-like” and “oblate-like” nuclei

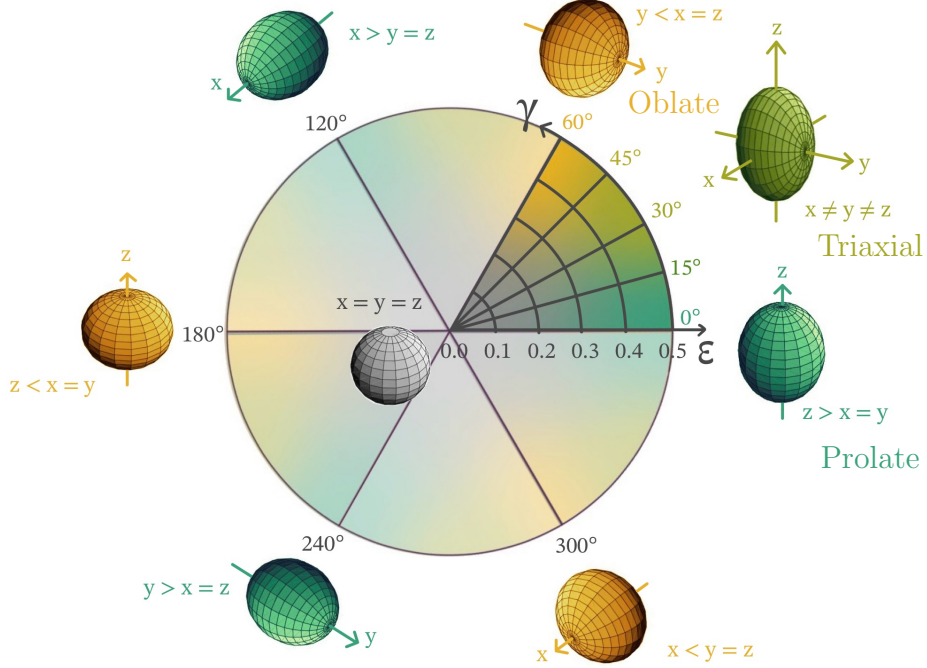


Figure 3: An illustration of the polar (ε, γ) parametrisation of deformation. The shaded sector ($\gamma = [0, 60]^\circ$) is sufficient to describe all deformations. The remaining sectors correspond to relabelling the principal axes (x, y, z) .

with $\gamma < 30^\circ$ and $\gamma > 30^\circ$ respectively. A value of $\gamma = 30^\circ$ corresponds to maximum triaxiality (neither prolate nor oblate), and a value of $\gamma = 0^\circ$ or 60° corresponds to axial symmetry (perfectly prolate or oblate, with no triaxiality).

Direct experimental evidence for triaxiality is difficult to obtain [10], however it is associated with several observable phenomena; exotic rotations such as wobbling motion and chiral rotation are characteristic [11] and can serve as evidence of triaxial deformation. The potential energy surface minimum is often very shallow in the triaxial direction [12], leading to γ -softness, where the triaxiality fluctuates over time [13].

An alternative approach is to identify patterns in existing experimental data, using commonly measured properties such as the ground state spin-parity (I^π), magnetic dipole moment (μ_D), and excitation energies, which may serve as evidence for triaxiality. The code employed in this project is the Particle-plus-Triaxial-Rotor Model (PTRM) described originally by Larsson, Leander, and Ragnarsson in 1978 [14]. These codes are written in the FORTRAN 77 programming language. Since the development of these programmes, they have undergone only a few minor modifications in 1988 [15]. More sophisticated models for the describing triaxiality have since been developed, which have the potential to yield more accurate results. Examples include the triaxial projected shell method and the generator coordinate method [16], as well as the use of covariant density functional theory to apply relativistic models to the particle-plus-triaxial-rotor [11].

Despite the availability of these newer models, the PTRM remains in use to this day. It has been utilised in a wide range of recent publications [17–26] for the interpreta-

tion of experimental data. Applications have included assigning spin states, identifying wobbling bands, deducing nuclear deformations, shape changes, shape coexistence, and several more phenomena. The continued use of these PTRM codes shows their significance to the ongoing study of nuclear shapes, and the importance of understanding their strengths and limitations.

The constraints of computational power at the time of development necessitated a relatively simple model. However, this simplicity has the advantage of quick calculation times on modern computers (less than 0.1 seconds per program), which allows for many calculations to be performed without the need for huge computing power. This enables a new application of these codes: it is now possible to efficiently study a large region of deformation space in a single set of calculations.

This project aims to explore the influence of triaxiality on the spectroscopic properties of a range of nuclei in the lead region. Calculations will be performed across the full range of triaxiality, as well as quadrupole deformations ranging from spherical to well-deformed. Results will be compared with a range of nuclear properties, including the ground state magnetic dipole moment, and the spins and energies of excited states, to infer ground state deformations in support or contradiction of existing predictions.

The following sections will describe the theoretical basis of the calculations (Sec. 2), and the approach taken to automate the computation (Sec. 3). This will be followed by a discussion of the results obtained (Sec. 4), with some comments on the consequences for future work.

2 Theoretical Model

The PTRM describes the rotational excitation spectrum of a deformed odd-mass nucleus. It couples single particle states from the Deformed Shell Model with rotational excitations from the Rigid-Rotor Model. It additionally accounts for the pairing interaction between nucleons.

The total Hamiltonian is given by [15]:

$$\hat{H}_{PTRM} = \hat{H}_{odd} + \hat{H}_{rot} + \hat{H}_{inter} + \hat{H}_{pair} \quad (5)$$

where $\hat{H}_{odd}$ gives the energy of the odd particle in the triaxial modified oscillator potential, as detailed in Sec. 2.1. Sec. 2.2 describes the core rigid rotor energy $\hat{H}_{rot}$, and the interaction energy between the core and the odd particle, given by $\hat{H}_{inter}$. The pairing energy $\hat{H}_{pair}$ is described in Sec. 2.3. Following this, the electromagnetic transition strengths and rates can be calculated, and the spins, energies, electric quadrupole moments, and magnetic dipole moments of the ground and excited states are obtained.

2.1 Single Particle States

The simplest quantum mechanical model of a nucleus assumes spherical symmetry. This is a microscopic picture, in which nucleons fill orbitals in order of increasing energy. These orbital energy levels are the eigenstates of the Spherical Shell Model Hamiltonian [9], given by:

$$\hat{H}_{sph} = -\frac{\hbar^2}{2m}\nabla^2 + V \quad (6)$$

where $\hbar$ is the reduced Planck constant, m is the nucleon mass, ∇ is the gradient operator, and V is the nuclear potential.

Unlike the atomic case, there is no central potential in the nucleus. Instead, a collective potential is assumed, generated by the nucleons themselves. There are various models to describe this potential; a common choice is the Harmonic Oscillator for the simplicity of its analytical solutions.

$$V(r) = \frac{1}{2}m(\overset{\circ}{\omega}_0)^2 r^2 \quad (7)$$

Here, r is the radial coordinate ($r^2 = x^2 + y^2 + z^2$), and $\overset{\circ}{\omega}_0$ is the spherical oscillator frequency. This is a constant defined for a given nucleus, separately for protons and neutrons due differences in their distributions [9].

Several modifications to this Hamiltonian are required. Firstly, deformation can be introduced by adapting $(\overset{\circ}{\omega}_0)^2 r^2$ to the case where $\omega_x \neq \omega_y \neq \omega_z$. We now relate the three oscillator frequencies in terms of quadrupole deformation (ε) and triaxiality (γ), as described in Eq. 4b. By expressing the triaxial deformed harmonic potential in terms of spherical harmonics (Y_{lm}) and stretched coordinates we simplify the extension to higher order multipole deformations (hexadecapole deformation ε_4 , and beyond) [9]:

$$V_{tri}(\rho) \approx \frac{1}{2}\hbar\omega_0\rho^2 \left[1 - \frac{2}{3}\varepsilon\sqrt{\frac{4\pi}{5}} \left(\cos(\gamma Y_{20}) - \frac{\sin(\gamma)}{\sqrt{2}}(Y_{22} + Y_{2-2}) \right) + \mathcal{O}(\varepsilon_4) \right] \quad (8)$$

where ρ is the stretched radial coordinate.

Applying a deformation lifts the energy degeneracy which previously held all orbitals within a principle shell ($\mathcal{N}$) of the oscillator to be equal. For axial symmetry ($\gamma = 0^\circ$ or 60°), states are split based on the number of oscillator quanta in the z -direction ($n_z = 0, 1, \dots, \mathcal{N}$) [9]. This effect is depicted in Fig. 4.

Secondly, the base of the harmonic oscillator potential is flattened with the addition of a centrifugal potential, $(-\kappa\mu\hbar\omega_0\vec{\ell} \cdot \vec{\ell})$, where $\vec{\ell}$ is the orbital angular momentum. The stronger this term is, the less diffuse the surface of the nucleus. This also reproduces the constant matter density within the nucleus, due to the incompressibility of nuclear matter. Microscopically, the effect of this term is to depress high- ℓ orbitals. The constants κ and μ are both experimentally determined, and defined separately for protons and neutrons [9]. The centrifugal term is further adjusted by subtracting its average value over all principle shells, $\langle \vec{\ell} \cdot \vec{\ell} \rangle_{\mathcal{N}}$. This avoids excessive compression of energy levels [9], such that the average energy of the levels is not affected, only their relative spacing.

Finally, the spin-orbit coupling effect between the orbital angular momentum and the intrinsic spin ($\vec{s}$) of a nucleon must be accounted for. Without this, the shell structure of the nuclear energy levels cannot be reproduced, therefore we add another term $(-2\kappa\hbar\omega_0\vec{\ell} \cdot \vec{s})$ to the potential (see Fig. 4). The full triaxial modified oscillator Hamiltonian, which gives the energy of a single particle in an orbital defined by $(\vec{\ell}, \vec{s})$, is now given by:

$$\hat{H}_{tri}(\vec{\ell}, \vec{s}) = -\frac{\hbar^2}{2m}\nabla^2 + V_{tri}(\rho, \vec{\ell}, \vec{s}) \quad (9)$$

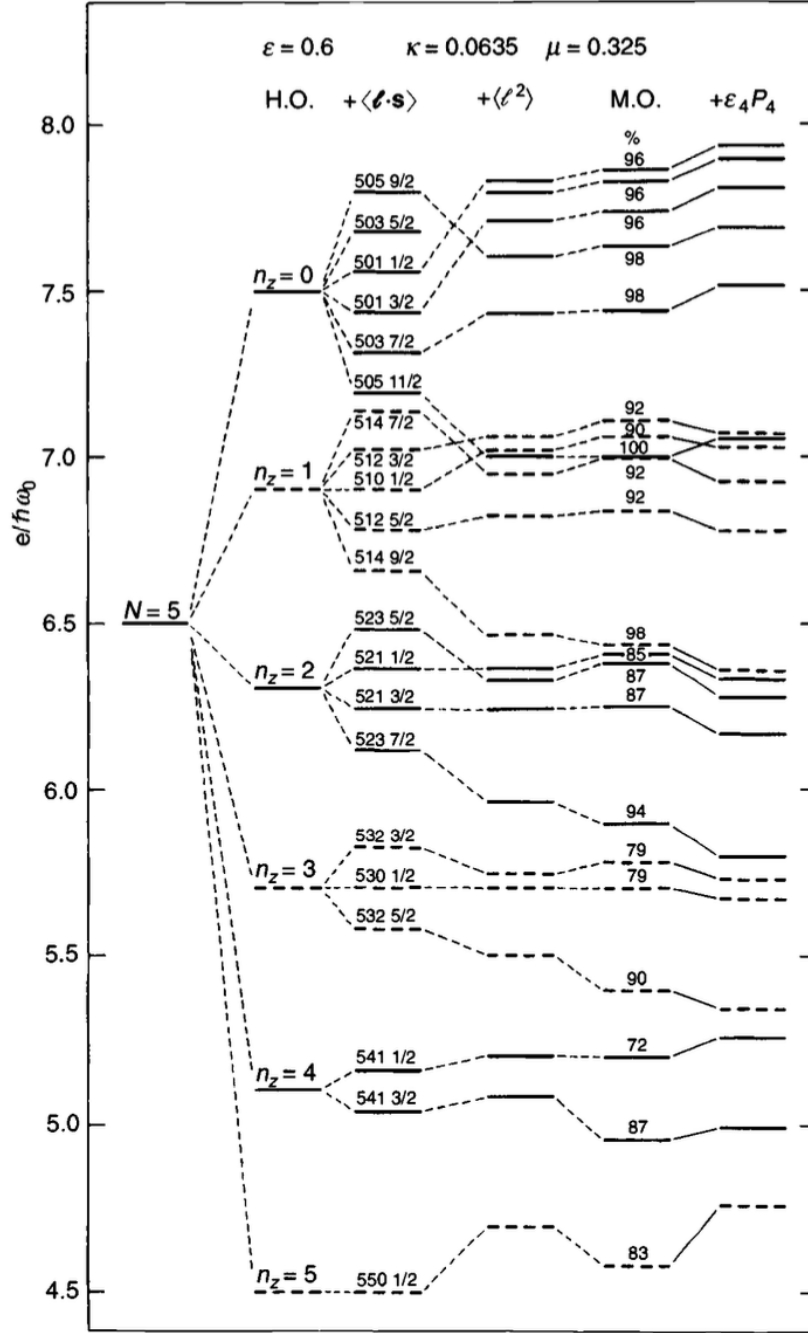


Figure 4: An illustration of the contributions to the single particle energy levels in an axially symmetric potential. Initially the energy levels of a principle shell ($\mathcal{N} = 5$ in this example) are degenerate. A deformation of $\varepsilon = 0.6$ is applied, splitting the levels according to n_z , the number of nodes along the z axis of the wavefunction. Then the spin-orbit and centrifugal terms are added, and an exact diagonalisation transforms the deformed harmonic oscillator (H.O.) potential to the modified oscillator (M.O.) potential of the Nilsson model. The purity of each state with respect to its $[\mathcal{N}n_z\Lambda]$ labelling is noted. Finally a higher order deformation, $\varepsilon_4 = 0.07$, is applied. This figure is taken from Fig. 8.2 of [9].

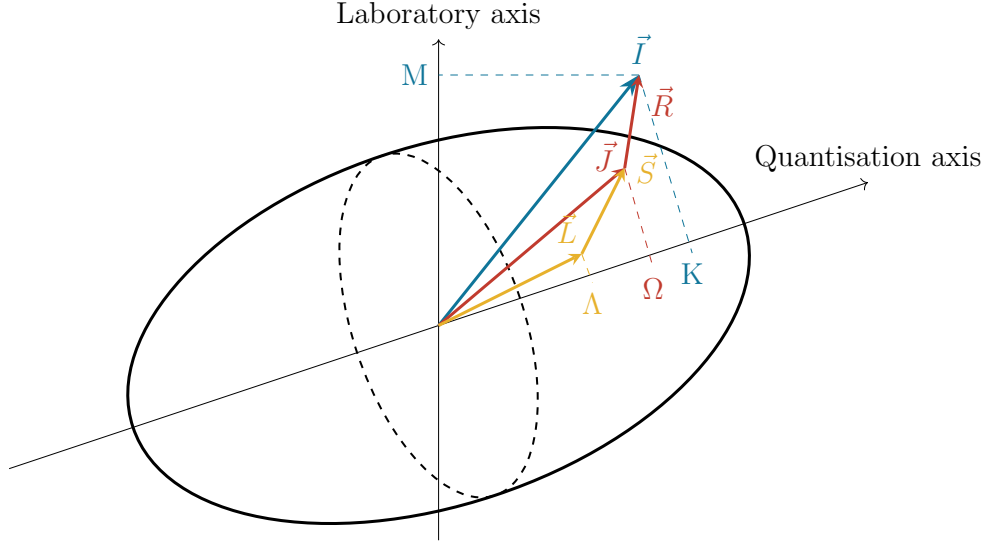


Figure 5: The PTRM quantum numbers. The total orbital angular momentum and intrinsic spin (of all the nucleons together) are drawn in yellow. Their sum is $\vec{J}$, which couples with the rotational angular momentum $\vec{R}$ (both drawn in red) to give the total angular momentum $\vec{I}$, drawn in blue. Projections onto the axes are also indicated.

where:

$$V_{tri}(\rho, \vec{\ell}, \vec{s}) \approx \frac{1}{2} \hbar \omega_0 \rho^2 \left[1 - \frac{2}{3} \sqrt{\frac{4\pi}{5}} \varepsilon \left(\cos(\gamma Y_{20}) - \frac{\sin(\gamma)}{\sqrt{2}} (Y_{22} + Y_{2-2}) \right) \right] - \kappa \mu \hbar \omega_0 (\vec{\ell} \cdot \vec{\ell} - \langle \vec{\ell} \cdot \vec{\ell} \rangle_{\mathcal{N}}) - 2 \kappa \hbar \omega_0 \vec{\ell} \cdot \vec{s} \quad (10)$$

These adjustments comprise the Deformed Modified Oscillator potential, which has been found to produce satisfactory results comparable with the Wood-Saxon potential [27]. The potential is now a function of $\vec{\ell}$ and $\vec{s}$ in addition to ρ (i.e. different orbitals experience different effective potentials). This further lifts the degeneracy of the energy levels, transforming from the deformed harmonic oscillator to the modified oscillator, as shown in Fig. 4.

In the special case where $\gamma = 0^\circ$ or 60° (i.e. the Nilsson single particle model [8] for axially symmetric deformations) the levels are split into projections (Ω) of their total angular momentum ($\vec{j} = \vec{\ell} + \vec{s}$) onto the symmetry axis (see Fig. 5). The result is that a maximum of only two nucleons (of each kind) can fill each energy level.

In prolate nuclei, where the symmetry axis is longer than the perpendicular axes, the large- Ω levels rise in energy. This is because a larger component of the angular momentum is parallel to the symmetry axis (see Fig. 5). The particle moves in a plane perpendicular to this vector, so it is confined to a narrower and steeper potential, which raises the energy levels. An analogous argument causes the low- Ω levels to rise in energy in oblate nuclei.

The Nilsson levels are labelled as $\Omega^\pi[\mathcal{N}n_z\Lambda]$ (see Fig. 4), with parity $\pi = (-1)^\ell$, and “asymptotic” quantum numbers $[\mathcal{N}n_z\Lambda]$, which are only truly valid at infinite deformation. Λ is the projection of $\vec{\ell}$ onto the symmetry axis (as shown in Fig. 5). At finite deformation, mixing between orbitals results in non-zero off-diagonal matrix

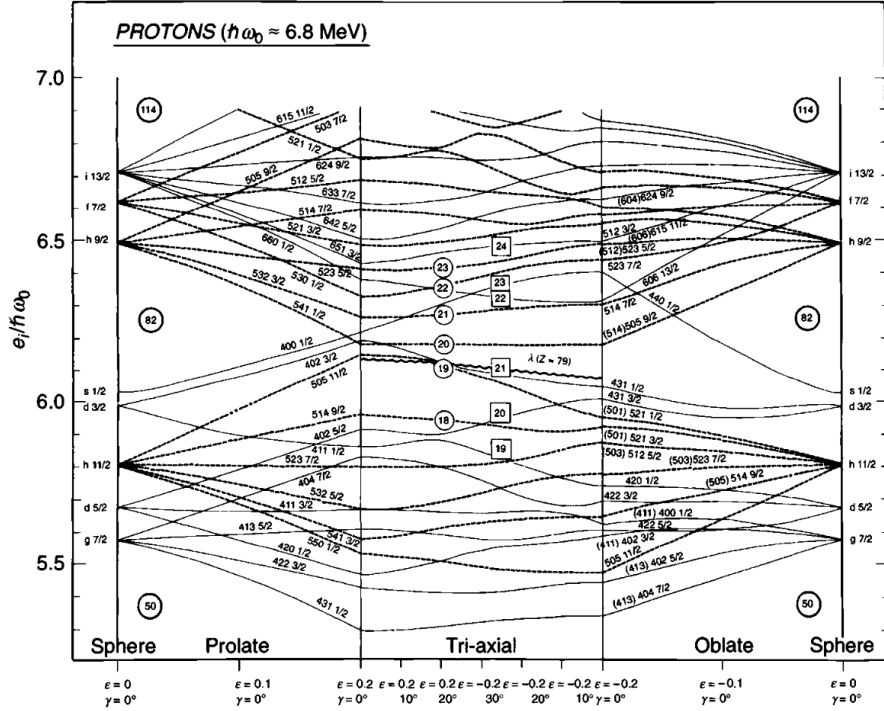


Figure 6: Variation in energy levels as deformation increases from spherical to prolate ($\epsilon = 0.2$), before transitioning to oblate ($\epsilon = -0.2$) through the triaxial parameter (γ), and returning to spherical ($\epsilon = 0$). In the axially-symmetric regions, the levels are labelled with $\Omega^\pi [\mathcal{N} n_z \Lambda]$. There is some mixing between levels of the same Ω^π here (indicated by labels in brackets). However in the triaxial region, there is strong mixing between all levels of the same parity (π). This Figure is taken from Fig. 8.7 of [9].

elements, and the states are not pure (as shown in Fig. 4). However in practice, the off-diagonal terms remain small enough that these are essentially valid quantum numbers for both small and intermediate quadrupole deformations [9]. Unlike the asymptotic quantum numbers, $[\mathcal{N} n_z \Lambda]$, the parity and spin-projection (Ω^π) are good quantum numbers at all axially-symmetric deformations.

In the triaxial case, parity remains a good quantum number (if only even-multipole deformations are included) however Ω is no longer conserved, and $[\mathcal{N} n_z \Lambda]$ are no longer reliable approximations. Instead, the wavefunctions are expanded in the basis $|\mathcal{N} \ell \vec{I} \Omega\rangle$

$$|\nu\rangle = \sum_{\mathcal{N} \vec{I} \Omega} C_{\mathcal{N} \vec{I} \Omega}^\nu |\mathcal{N} \ell \vec{I} \Omega\rangle \quad (11)$$

where $C_{\mathcal{N} \vec{I} \Omega}^\nu$ are the expansion coefficients [15]. As triaxiality increases, the mixing of orbitals becomes significant. This can be seen in Fig. 6, where the triaxial region in the centre bridges between states of different quantum numbers on the prolate and oblate sides of the deformation space. A numerical solution for the single-particle energy levels can now be obtained by expanding the wavefunction in this basis, truncated to low- $\mathcal{N}$, and diagonalising this Hamiltonian.

2.2 Collective Rotational Excitations

A perfectly spherical nucleus cannot rotate, as there can be no rotation around a symmetry axis of a quantum body (this would only result in an unobservable phase shift of the wavefunction) [9]. Therefore, deformation opens up rotational degrees of freedom. Classically, the rotational kinetic energy of a rigid body is given by:

$$E_{rot} = \frac{L^2}{2\mathcal{I}} \quad (12)$$

where $\mathcal{I}$ is the moment of inertia, and the angular momentum $L = \mathcal{I}\omega_v$ for angular velocity ω_v . Applying quantum mechanics, this becomes:

$$\hat{H}_{rot} = \frac{\hat{R}^2}{2\mathcal{I}} \quad (13a)$$

$$E_{rot} = \frac{\hbar^2}{2\mathcal{I}} R(R+1) \quad (13b)$$

where $\hat{R}$ is the rotational angular momentum operator. In even-even nuclei, its eigenvalues are even-integers, $R = [0, 2, 4, 6, \dots]\hbar$. This rigid-rotor model predicts a structured spacing of the energy levels in a rotational band. In even-even nuclei, where the ground state is $\vec{J}^\pi = 0^+$, the total angular momentum ($\vec{I} = \vec{R} + \vec{J}$) is equal to the rotational angular momentum ($\vec{I} = \vec{R}$). We therefore expect to see a rotational band with energy levels at $E_{rot} = [0, 6, 20, 42, \dots]\frac{\hbar^2}{2\mathcal{I}}$.

So far, the collective rotational energy levels have been considered independently from the single particle energy levels of Sec. 2.1. In odd-mass nuclei, the PTRM considers the odd particle to be coupled to a rotating even-even core. In this configuration, both odd and even integers $\vec{R}$ are allowed. Since $\vec{J}$ is a half-integer for odd-mass nuclei, this results in the half-integer $\vec{I}$ states observed in experimental spectra.

The total spin, $\vec{I}$, now includes both non-zero $\vec{J}$ and $\vec{R}$. Furthermore, in the triaxial case the angular momenta and moments of inertia are defined separately for the three principal axes $k = (x, y, z)$. This leads to the triaxial rotational Hamiltonian [14]:

$$\hat{H}_{rot} = \sum_k \left[\frac{\hat{R}_k^2}{2\mathcal{I}_k} \right] \quad (14)$$

Expanding ($\vec{R} = \vec{I} - \vec{J}$) we get:

$$\hat{H}_{rot} = \sum_k \frac{\hat{I}_k^2}{2\mathcal{I}_k} - \sum_k \frac{\hat{I}_k \hat{J}_k}{2\mathcal{I}_k} + \sum_k \frac{\hat{J}_k^2}{2\mathcal{I}_k} \quad (15)$$

The three terms in Eq. 15 correspond to the core rotational kinetic energy ($\hat{H}_{core}$), modified by a Coriolis term and a recoil term, which together form $\hat{H}_{inter}$ [14]. Now the first three terms of the PTRM Hamiltonian (Eq. 5) have been accounted for.

The wavefunctions of these particle-plus-rotor coupled energy levels are expanded in the basis $|\vec{I}MK\nu\rangle$ [15], where $\vec{I}$ is the total spin, M is its projection onto the laboratory axis, K is its projection onto the intrinsic quantisation axis, and ν is the single particle orbital (see Eq. 11 and Fig. 5).

In the axially symmetric case ($\gamma = 0^\circ$), $\vec{R}$ is strictly perpendicular to the symmetry axis, such that there is no rotation around the symmetry axis. K is therefore restricted to Ω . As $\vec{R}$ increases in a rotational band, $\vec{I}$ increases in integer steps, while K (a preserved quantum number under axial symmetry) is constant [9]. However in the triaxial case, K is not a well-defined quantum number, and K states of the same $\vec{I}$ mix.

For small $\vec{I}$, the internal structure of the core is not significantly affected by the rotation, and the Coriolis and recoil terms ($\hat{H}_{inter}$) can be considered perturbations to the core rotor energy ($\hat{H}_{rot}$). This is known as the *strong coupling limit* [9]. In this limit, the basis that diagonalises the PTRM Hamiltonian (Eq. 5), excluding the pairing term, is the “strong coupling basis”. The use of this basis is intended to allow for faster numerical convergence of the calculations, at any deformation, without requiring a large set of single particle levels to be involved [14].

Making these calculations requires input of the moment of inertia, $\mathcal{I}$. The effective value of this parameter can be determined approximately from the excitation energy of the lowest 2^+ state in the core rotor ($E(2^+)$), since from the even-even rigid rotor model (Eq. 13b) we expect:

$$\mathcal{I} = 6 \frac{\hbar^2}{2E(2^+)} \quad (16)$$

This total moment of inertia is then distributed between $k = (x, y, z) = (1, 2, 3)$ according to a hydrodynamical relation [14]:

$$\mathcal{I}_k = \frac{4}{3} \mathcal{I} \sin^2 \left(\gamma + \frac{2\pi}{3}k \right) \quad (17)$$

The value of $E(2^+)$ for this calculation can be found from the experimental spectrum of the even-even core. However, this is not always a reliable means of determining the moment of inertia of the rotor. The core can always be defined in two ways, such that the nucleus is either the core-plus-particle, or the core-plus-hole. These two choices of the core do not necessarily have a consistent $E(2^+)$ value. Furthermore, the addition or removal of the single particle can have a significant influence on the moment of inertia of the nucleus, and so it may be necessary to tune the *effective* value of $E(2^+)$ to obtain an improved agreement of calculated results to experimental data [14].

Moreover, the PTRM assumes the core to be rigid with a fixed moment of inertia, independent of rotation. For this reason, the rigid rotor model increasingly overestimates the experimental energy levels at higher spins [28]. Further problems can arise in nuclei without a strongly preferred deformation. The rigid core approximation cannot account for effects such as γ -softness, therefore it cannot be used to model vibrations.

2.3 Pairing Energy

Nucleons filling these energy levels form spin pairs, resulting in a pairing energy term, $\hat{H}_{pair}$. In the strong coupling basis (described in Sec. 2.2), $\hat{H}_{pair}$ is diagonal [15]. This is calculated using the Bardeen–Cooper–Schrieffer (BCS) method applied to a subset of the single particle levels [9, 14, 29]. The BCS method was originally devised in the context of Cooper pairs in superconductors [30].

Firstly, the pairing strength parameter (G) must be determined. One possible formulation is given by:

$$G = \frac{1}{A} \left(g_0 \mp g_1 \frac{N-Z}{A} \right) \text{MeV} \begin{cases} - & \text{for neutrons} \\ + & \text{for protons} \end{cases} \quad (18)$$

This is defined separately for protons and neutrons. The parameters g_0 and g_1 are fitted to experimental data for a given region of the nuclear chart. In the lead region, the standard values are $g_0 = 19.2$ MeV, $g_1 = 7.4$ MeV [9].

Secondly, two coupled non-linear equations are solved together [29] to obtain the pairing gap (Δ) and BCS Fermi energy (E_f):

$$n = 2 \sum_i V_i^2 \quad (19)$$

$$\frac{2}{G} = \sum_i \frac{1}{E_i} \quad (20)$$

In these equations, the sums run over a set of orbitals, i , chosen from the single particle orbitals. The standard number of orbitals to include is $\sqrt{15n}$ either side of the Fermi level, where n is the number of nucleons (either protons or neutrons depending on the odd particle). E_i is the set of quasiparticle energies, calculated with:

$$E_i = \sqrt{(\epsilon_i - E_f)^2 + \Delta^2} \quad (21)$$

where ϵ_i are the set of single particle orbital energies. V_i^2 are the corresponding occupation probabilities (i.e. the probability for orbital i to be occupied by a pair of nucleons), which can be calculated with:

$$V_i^2 = \frac{1}{2} \left[1 - \frac{\epsilon_i - E_f}{\sqrt{(\epsilon_i - E_f)^2 + \Delta^2}} \right] \quad (22)$$

The related quantity U_i^2 expresses the probability that orbital i is *not* occupied:

$$U_i^2 = \frac{1}{2} \left[1 + \frac{\epsilon_i - E_f}{\sqrt{(\epsilon_i - E_f)^2 + \Delta^2}} \right] \quad (23)$$

U_i and V_i are normalised on the diagonal by the condition $U_i^2 + V_i^2 = 1$. The pairing occupation factor ($U_i U_{i'} + V_i V_{i'}$) is calculated for each single particle orbital in the input subset, and their energies are replaced with quasiparticle energies (Eq. 21). This pairing factor multiplies the matrix elements of the PTRM Hamiltonian (Eq. 5), diagonalised in the single quasiparticle strong coupling basis; thus the pairing interaction is accounted for.

3 Computational Method

The codes used in this work are the series of three Modified-Oscillator-based programmes (GAMPN, ASYRMO, and PROBAMO) in the collection of Particle-plus-Triaxial-Rotor Fortran codes presented at the Risö Hands-On Nuclear Structure Theory Workshop (June 1992), which were adapted from the original Oak Ridge versions of the

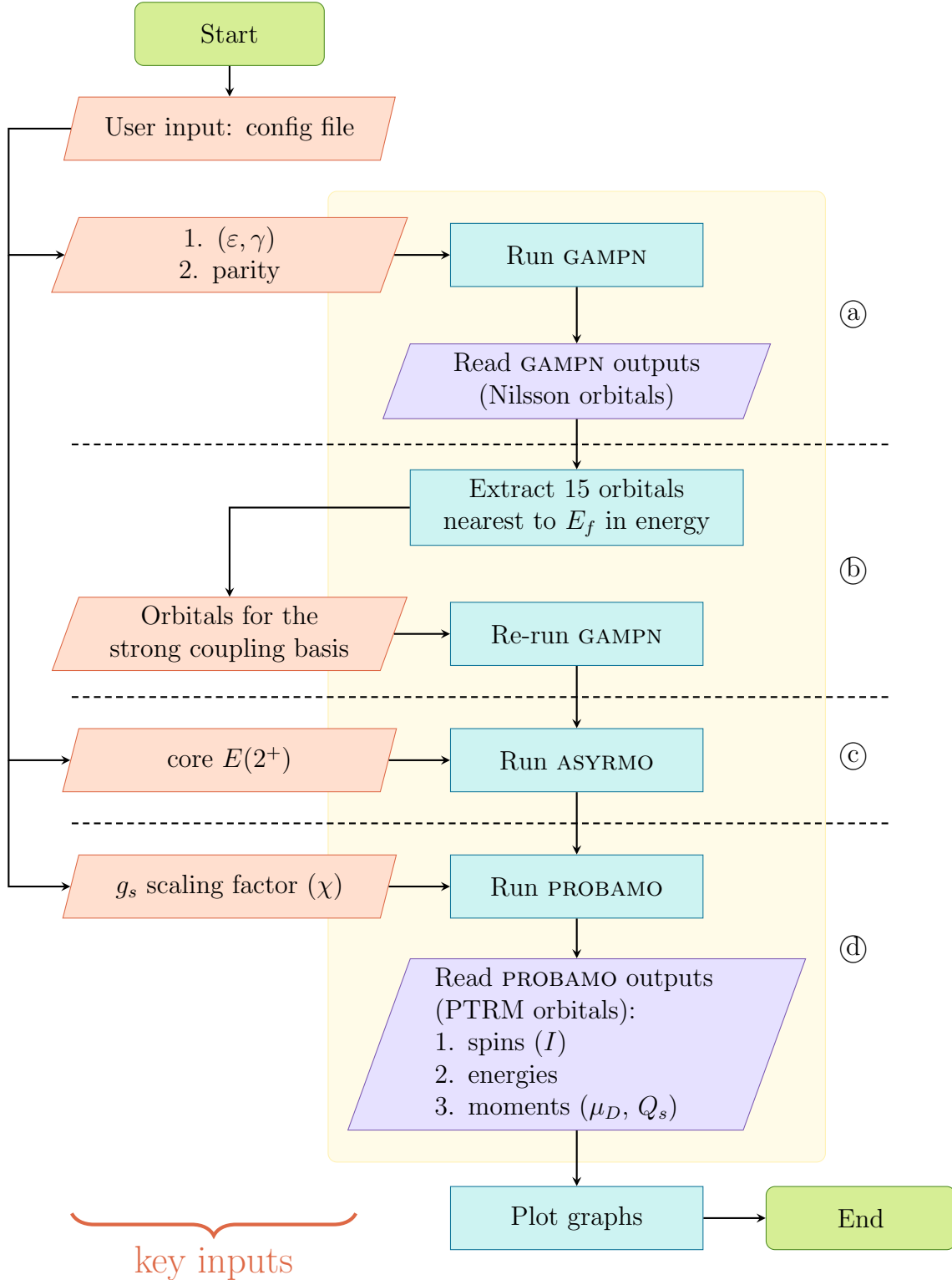


Figure 7: A flowchart illustrating the algorithm used to automate the codes in Python. The four main stages labelled ①-④ are described in the main text. The blue boxes represent the primary processes. The key inputs are shown in red. The key outputs occur in two parts, shown in purple, where the results of the calculations are extracted from the output files.

previous year [31]. A Python interface has been developed² to automate calculations using these programmes, responding dynamically to the output of each stage before managing inputs to the next. The usability of the original programmes is enhanced with a simplified input process, graphic outputs, and clear error messages for known errors. The implementation is efficient, robust and flexible, capable of utilising the full computing power of a machine of any size, and enabling calculations on scales ranging from a single deformation to more than a thousand.

Figure 7 gives an overview of the process used to automate the running of these programmes in four main stages, labelled (a)-(d). The yellow box on the figure highlights the processes which must be run in a loop over each deformation before moving on to the next step. For example, in a small set of three different deformations, a separate GAMPN input file must be written for each, and GAMPN must be run three times. Then the code will move on to reading the three output files, and selecting the three sets of orbitals for the construction of the strong coupling basis, *etc.*

For multiple calculations, the computing time scales linearly with the number of deformations, where a typical calculation at a single deformation takes ~ 0.1 seconds. Therefore, efficient implementation is essential. For data sets with more than 20 points, the set is divided up into P evenly-sized batches, where P is the number of computer cores to utilise. A shell script is generated for each batch, containing commands to run each file in sequence and save the outputs. These shell scripts are started asynchronously as subprocesses using the `Popen` class of Python’s `subprocess` module. This allows the heaviest computational workload to be run in parallel, with the opportunity to scale to any number of processors on larger machines. For a set of 820 data points, increasing the number of cores from 1 to 8 results in typical runtime decreasing from 63 seconds to 15 seconds; a $4.1\times$ speedup.

All of the user input occurs at the start, via a “config” file. There are many input parameters which can be used to tune the calculations, and these are detailed in [31]. The parameters of primary importance will be described where relevant in the remainder of this section, which elaborates on the contents of stages (a)-(d).

In part (a) of Fig. 7, the first program (GAMPN) solves the Schrödinger equation for the odd particle in the deformed potential, obtaining the single particle levels as in Sec. 2.1. Four different modes are supported for the input of ε and γ deformations (see Fig. 8):

1. A single point, (ε, γ) .
2. A range of ε with fixed γ .
3. A range of γ with fixed ε .
4. The maximum value of ε to be tested in a mesh of points covering the full range of $\gamma = [0, 60]^\circ$ and $\varepsilon = [0, \varepsilon_{max}]$. In this case, the total number of data points is given by $\frac{1}{2}n(n+1)$, where n is the number of γ values to test at ε_{max} (see Fig. 8d). This creates an even distribution of data points in polar (ε, γ) space.

This flexibility is convenient for performing multiple tests at different scales. By convention, γ is typically input in the range $[0, 60]^\circ$ and $\varepsilon > 0$. In option 4, this is required,

²The code developed in this work can be found in a GitHub repository at <https://github.com/katyrr/Exploring-Triaxiality-in-the-Lead-Region>.

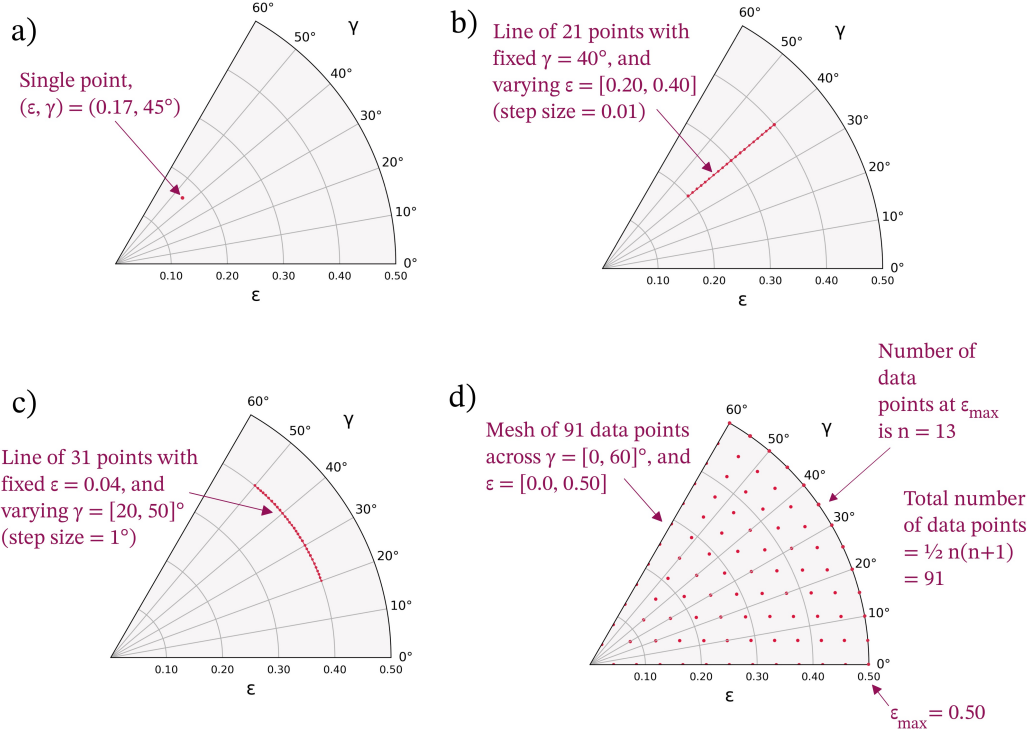


Figure 8: Example deformation inputs: (a) a single point, (b) a range of ϵ with fixed γ , (c) a range of γ with fixed ϵ , and (d) a mesh covering $\gamma = [0, 60]^\circ$, $\epsilon = [0.0, 0.50]$. In (d) this example shows 13 data points at $\epsilon_{\max}$ in the range $\gamma = [0, 60]^\circ$, therefore it contains 91 data points in total.

although in all other cases any convention can be used. In practice, the calculation cannot be performed at an exactly spherical deformation ($\epsilon = 0$) as the PTRM applies only to deformed nuclei, therefore $\epsilon = 0.001$ is used as an approximation. Higher order multipole deformations (ϵ_4, ϵ_6) can be defined in terms of spherical harmonics (Y_{lm}) with $l = 4, 6$, analogous to the quadrupole ($l = 2$) deformation (Eq. 2). However, these only constitute minor perturbations to the shape, and are therefore set to zero in these calculations.

The nucleus to be studied is described by its proton and mass numbers, Z and A . The number of neutrons is then $N = A - Z$. The Fermi level is calculated as $\frac{N+1}{2}$ or $\frac{Z+1}{2}$ depending on whether the odd particle is a neutron or a proton. This single particle level is identified in each of the GAMPN output files, and the Fermi energy is recorded.

The desired parity of the output spectrum must be specified as either positive or negative at this stage; only one can be calculated at a time. Therefore, states of different parities cannot be compared quantitatively, as there is no mixing between them, and there is an overall relative depression of negative parity levels, which may be caused by the treatment of pairing effects in this model [14].

In part (b), the fifteen orbitals (of the input parity) that lie closest in energy to the Fermi level are identified at each data point. Then GAMPN is run again, now using these fifteen orbitals as input to construct the strong coupling basis. All single particle matrix elements required for construction of the PTRM Hamiltonian and electromag-

netic transition matrices are calculated at this point, and stored in a separate file which will be read into ASYRMO and PROBAMO. The use of the new set of orbitals has no impact on the single particle energy levels, therefore there is no need to read the output file again.

In part   the second program, ASYRMO, applies the BCS approximation to calculate pairing interactions [9], as described in Sec. 2.3. The PTRM Hamiltonian is then diagonalised in the single quasiparticle strong coupling basis. In order to do this, the effective energy of the lowest 2^+ excited state in the core ($E(2^+)$) must be input, so that the moment of inertia of the rigid core can be estimated. This is initially taken from the experimental spectrum of the even-even core (either the neighbour above or below the nucleus being studied).

In the event that a poor agreement to experimental data is obtained at all deformations tested, a range of $E(2^+)$ values can be tested to improve the results. The justification for this approach was discussed in Sec. 2.2. Another option is to estimate the value of $E(2^+)$ using Grodzin’s relation [9], which can be calculated dynamically by the script using:

$$E(2^+) \approx \frac{1100}{\varepsilon^2 A^{7/3}} \text{ MeV} \quad (24)$$

This may be a more appropriate method for nuclei near closed shells, where the core is generally near-spherical. In this case, the experimental low-lying 2^+ state is not typically rotational in character, therefore attempting to infer the effective core moment of inertia from this is not well-justified.

Low spin states ($I, K \leq \frac{13}{2}$) are calculated for up to five states of each spin. The pairing strength parameter is calculated with Eq. 18, using the standard values described in Sec. 2.3. Coriolis attenuation is not applied.

In part   the third program, PROBAMO, calculates the electromagnetic matrix elements. The diagonal elements are the static moments: the magnetic dipole moment (μ_D) and the spectroscopic electric quadrupole moment (Q_s). In the calculation of μ_D , a scaling factor $\chi = [0, 1]$ is applied to the free g -factor for intrinsic spin:

$$g_{s,\text{eff}} = \chi g_{s,\text{free}} \quad (25)$$

This accounts for effects such as core polarisation and the recoil effect [9], which the PTRM cannot model explicitly. The g -factor of the core is calculated as $g_R = \frac{Z}{A}$ [32]. The magnetic dipole moment of the ground state (units μ_N) is then given by [15]:

$$\mu_D = \sqrt{\frac{4\pi}{3}} \left\langle \vec{I}_0 \left| g_R \vec{R} + g_\ell \vec{\ell} + g_{s,\text{eff}} \vec{s} \right| \vec{I}_0 \right\rangle \quad (26)$$

where $|\vec{I}_0\rangle$ is the ground state wavefunction and $g_\ell = 1$ or 0 for an odd proton or neutron respectively.

From the output files, the spins and energies of all calculated states, as well as the magnetic dipole moment of the ground state, are recorded. These are plotted as functions of ε , γ . When varying both parameters in a mesh (as in Fig. 8(d)), a total of 820 data points was found to provide a satisfactory balance of resolution with time taken and memory usage. Trials of as many as 10011 data points were tested, however beyond 820 data points the resolution is no longer the main source of uncertainty, therefore further increasing the resolution did not impact the conclusions.

Table 1: Comparison of experimental properties of $^{207}_{82}\text{Pb}_{125}$ with calculated values at $\varepsilon = 0.001$, using $E(2^+) = 4833$ MeV. The experimental ground state magnetic dipole moment is taken from [33]. Key features of the excitation spectrum are listed in this table; the full experimental spectrum can be found in [34]. Uncertainties are quoted in brackets (see main text for details).

Low-lying negative-parity states	Energy relative to the $1/2^-$ ground state / keV	
	Experimental value	Calculated at $\varepsilon = 0.001$
$9/2^-$	3415.43(4)	3180(20)
$7/2^-$	2339.88(3)	3790(20)
$3/2^-$	897.92(2)	950(20)
$5/2^-$	569.70(2)	650(20)
Ground state property	Experimental value	Calculated at $\varepsilon = 0.001$
I^π	$1/2^-$	$1/2^-$
μ_D	0.59102(18) μ_N	0.6(1) μ_N

4 Results and Discussion

Three isotopes in the lead region are studied. $^{207}_{82}\text{Pb}_{125}$ is well known to be near-spherical in the ground state, and is therefore used to test the performance of the model. Calculations are then performed for $^{179}_{79}\text{Au}_{100}$ and $^{177}_{78}\text{Pt}_{99}$, to determine their deformations.

4.1 ^{207}Pb

$^{207}_{82}\text{Pb}_{125}$ is a neighbour of the doubly-magic isotope $^{208}_{82}\text{Pb}_{126}$ with one hole in the neutron shell closure $N = 126$. It is therefore expected to be near-spherical in the ground state ($\varepsilon \approx \gamma \approx 0$). Such a symmetric nucleus should be challenging for the PTRM to model, as there can be no rotation around a symmetry axis of a quantum body. Rotational excitations, if they are present at all, would be expected to occur at very high energies due to the small moment of inertia of the system (see Sec. 2.2).

Nonetheless, in the limit of $\varepsilon \rightarrow 0$, we expect the calculated properties of $^{207}_{82}\text{Pb}_{125}$ to approach their experimental values, which are summarised on the left of Table 1. Calculations were performed at $\varepsilon = 0.001$, with 120 data points across $\gamma = [0, 60]^\circ$. At this small value of ε the γ parameter is not very meaningful, however there is significant numerical noise when γ is close to 0° or 60° , and the higher values of γ provide some stability to the calculations. This can be seen in Fig. 9.

Grodzin’s relation is used to estimate the effective core $E(2^+)$ (see Sec. 2.2). At this

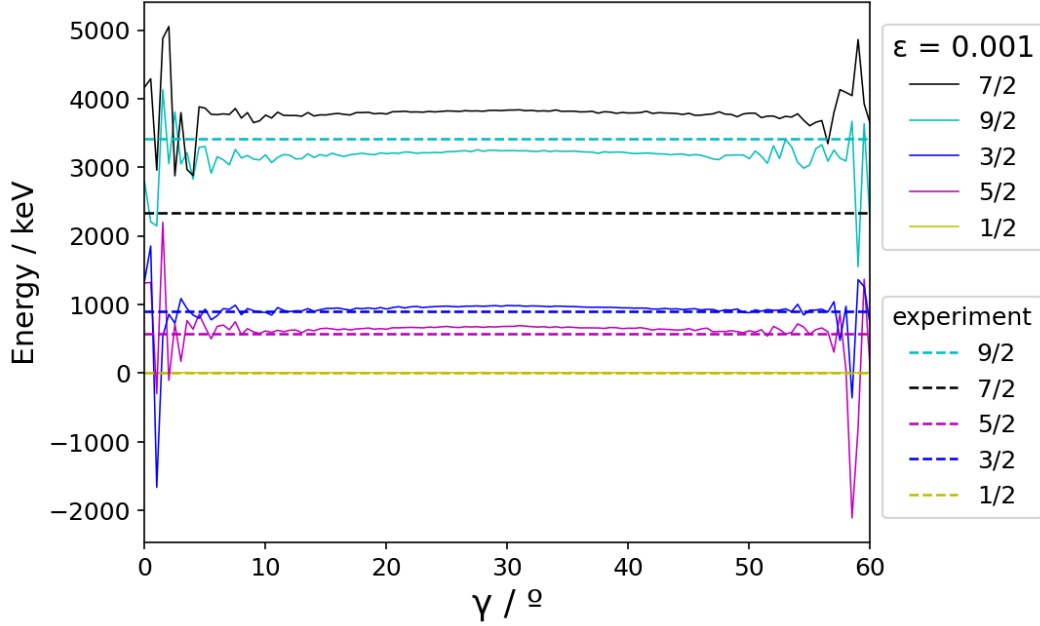


Figure 9: Calculated low-lying negative parity energy levels at $\varepsilon = 0.001$, relative to the lowest $\frac{1}{2}^-$ state (the expected ground state). $E(2^+) = 4833$ MeV (see main text). The dashed lines show the experimental values. The $I = \frac{7}{2}^-$ state is overestimated by the calculations (see main text), but the results are otherwise in agreement with experiment.

$\varepsilon = 0.001$ we find $E(2^+) = 4833$ MeV, as calculated with Eq. 24. Note that the units here are MeV; this is an extraordinarily large value, more than four orders of magnitude greater than is typical for a rotational nucleus, which reflects its reluctance to rotate. The result is that all of the calculated rotational energy levels become extremely large, far out of the range of the calculations. The five levels which remain at low energy are almost invariant with respect to the core rotation. These are the single particle excitations.

The experimental ground state spin ($I = \frac{1}{2}^-$) is reproduced, and the calculated $\frac{5}{2}^-$, $\frac{3}{2}^-$, and $\frac{9}{2}^-$ states show good agreement with the experimental values (see Fig. 9). These can all be understood by single particle excitations in the nuclear shell model (see Fig. 10). In the spherical shell model, we might expect the $\frac{7}{2}^-$ state to lie *above* the $\frac{9}{2}^-$ state, and this is where it appears in the calculations. Its location in the experimental spectrum has been attributed to coupling with the octupole vibration of the core [35]. However the PTRM does not model odd multipoles [14], therefore it cannot account for this coupling. This causes the observed discrepancy in the energy of the $\frac{7}{2}^-$ state, and any remaining differences may be caused by the relatively simple treatment of pairing in the PTRM [14].

The results of the calculations are summarised on the right of Table 1. The reported excitation energies are the mean values in the range $\gamma = [0, 60]^\circ$ (see Fig. 9). The uncertainty is estimated as the standard error in the mean. In the calculation of the magnetic dipole moment, no scaling factor has been applied to the intrinsic spin g -factor of the odd particle ($\chi=1.0$, see Sec. 3). This is because the single particle model for

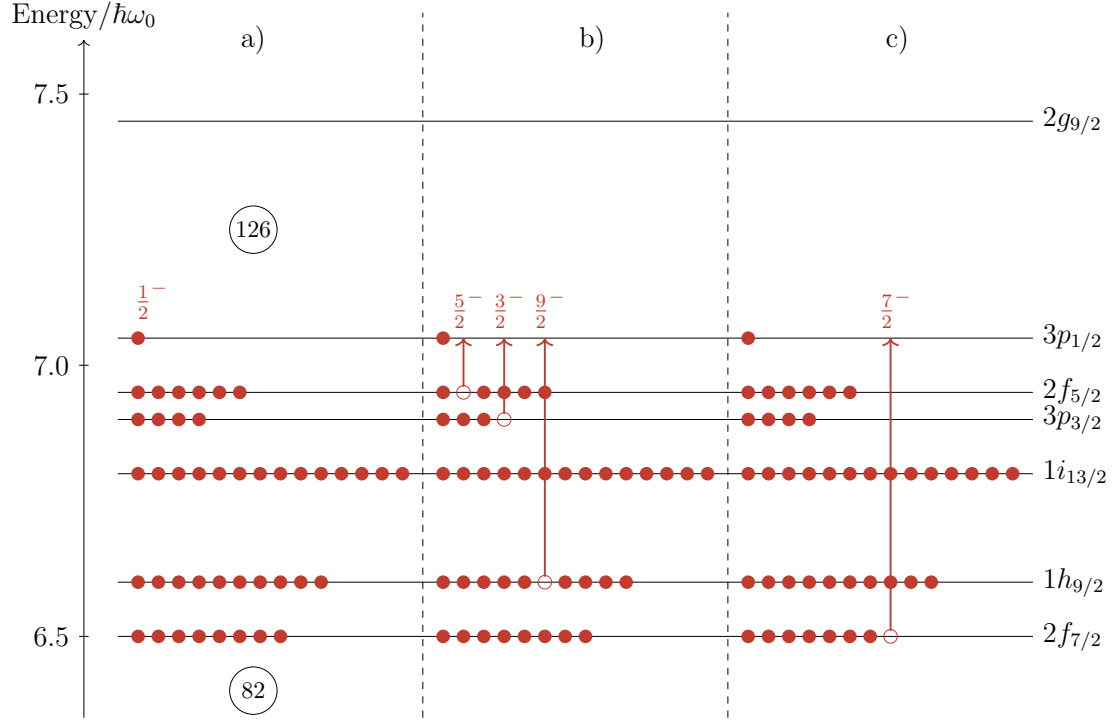


Figure 10: A schematic illustration of the energy levels in the spherical shell model. The neutron orbital occupancies of $^{207}\text{Pb}_{125}$ are shown in (a) the ground state, where a single particle in the $3p_{1/2}$ level results in the ground state $I^\pi = \frac{1}{2}^-$. In (b), three single particle excitations that lead to the observed negative-parity spectrum are indicated. In (c) the excitation which results in the calculated $\frac{7}{2}^-$ state is shown. The $1i_{13/2}$ orbital has positive parity and is therefore not involved in the negative parity spectrum.

calculating the magnetic dipole moment is a better approximation at closed shells [9], therefore the free value of the intrinsic spin g -factor can be used without adjustment.

These results illustrate the capability of the PTRM to model single particle excitations, which allows it to describe near-spherical nuclei despite the absence of rotations. However, the discrepancy between the experimental and calculated $\frac{7}{2}^-$ state demonstrates that caution is required when applying the PTRM to nuclei with octupole vibrations.

4.2 ^{179}Au

$^{179}_{79}\text{Au}_{100}$ is a neutron deficient isotope of the stable nucleus $^{197}_{79}\text{Au}_{118}$. It has an even number of neutrons near the neutron mid-shell, and an odd number of protons, leaving three holes in the $Z = 82$ proton shell closure. This allows for strong proton-neutron correlations which drive deformation. Furthermore, this nucleus and its neighbours are known candidates for triaxiality and shape coexistence [36, 37].

In the spherical shell model, the odd proton occupies level $2d_{3/2}$, which would imply a ground state of $I^\pi = \frac{3}{2}^+$. However, experimental measurements have assigned the ground state spin-parity as $I^\pi = \frac{1}{2}^+$ [36]. Measurements of μ_D and analysis of the associated g -factor implies that the odd proton wavefunction has a mixed configuration of $3s_{1/2}$ and $2d_{3/2}$ proton orbitals, indicating that this nucleus is deformed,

Table 2: Comparison of the experimental properties of $^{179}_{79}\text{Au}_{100}$ with the calculated values at $(\varepsilon, \gamma) = (0.12, 35^\circ)$ and at $(\varepsilon, \gamma) = (0.19, 35^\circ)$. The ground state properties are taken from [36]. Key features of the excitation spectrum are listed in this table; the full spectrum can be found in [18]. Uncertainties are quoted in brackets (see main text for details).

Low-lying positive-parity states	Energy relative to the $1/2^+$ ground state / keV		
	Experimental value	Calculated at $(0.12, 35^\circ)$	Calculated at $(0.19, 35^\circ)$
$13/2^+$	515(1)	800(100)	500(100)
$9/2^+$	498(1)	710(90)	400(40)
$3/2^+$	27(1)	11(7)	20(20)
Ground state property	Experimental value	Calculated at $(0.12, 35^\circ)$	Calculated at $(0.19, 35^\circ)$
I^π	$1/2^+$	$1/2^+$	$1/2^+$
μ_D	$1.01(5) \mu_N$	$1.0(5) \mu_N$	$0.6(3) \mu_N$

and may be triaxial [36]. Potential energy surface calculations suggest strong deformation with maximal triaxiality, identifying the potential energy surface minimum at $(\varepsilon, \gamma) = (0.150, 30.0^\circ)$ [38].

The ground state deformation of $^{179}_{79}\text{Au}_{100}$ can be deduced by fitting the spectroscopic properties calculated with the PTRM codes to the experimental data summarised on the left of Table 2. Calculations were run over a mesh of 820 data points in the range $\gamma = [0, 60]^\circ$ and $\varepsilon = [0, 0.5]$. The lowest 2^+ state of the neighbouring $^{178}_{78}\text{Pt}_{100}$ nucleus was used to define the core rotor spectrum (see Sec. 2.2), $E(2^+) = 0.17$ MeV [39]. This choice was found to provide a better agreement than the higher value from the other neighbour, $^{180}_{80}\text{Hg}_{100}$.

The calculated ground state spins and magnetic dipole moments are shown in Fig. 11. For μ_D , data points where the calculated value agrees with the experimental value within a tolerance of $\pm 0.2\mu_N$ are marked in red. This is considered a good agreement in this model [21], and distinguishes the best-fitting values approximately corresponding to the top tenth percentile.

There are three areas in Fig. 11(b) where the calculated ground state magnetic dipole moment reproduces the experimental value. However, only the group at $\varepsilon = [0.09, 0.15]$ and $\gamma = [30, 60]^\circ$ calculates the correct ground state spin (see Fig. 11(a)). This provides an initial suggestion of the region of interest.

A scaling factor of $\chi=0.7$ is applied to the free intrinsic spin g -factor. This is a typical value for odd-mass nuclei in the lead region [32]. However scaling factors in the range $[0.5, 0.8]$ are not uncommon, and the chosen value can impact the location of the

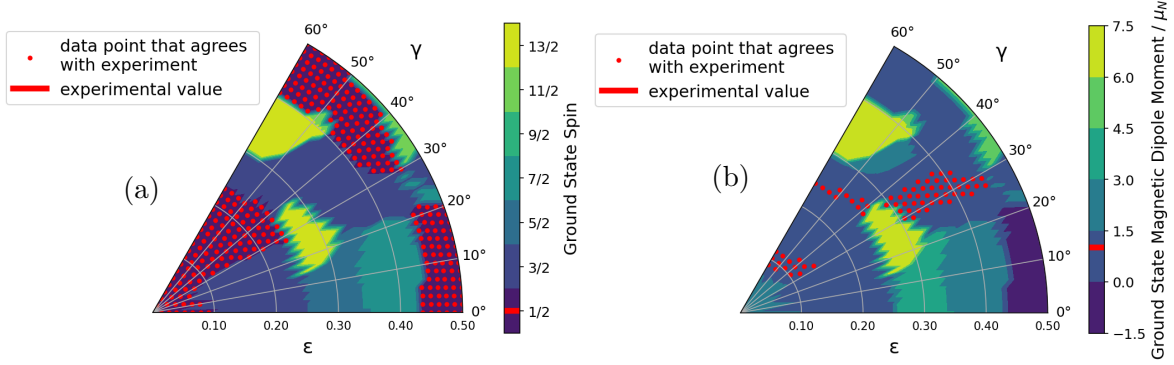


Figure 11: PTRM calculations of (a) the ground state spin, and (b) the ground state magnetic dipole moment of $^{179}\text{Au}_{100}$ across the deformation plane. The experimental value is marked in red on the colour bar. Data points where the calculated and experimental values agree (within $\pm 0.2 \mu_N$ for the magnetic dipole moment) are marked on the graph with red circles. These areas coincide on (a) and (b) at $(\epsilon, \gamma) \approx (0.12, 45^\circ)$, identifying an initial region of interest.

overlap region in Fig. 11. The bounds on the region of interest are therefore estimated to have an uncertainty of $\delta(\epsilon) = \pm 0.03$. On the other hand, γ is not sensitive to the chosen scaling factor in this case. The region of interest therefore expands to $\epsilon = [0.06, 0.18]$ and $\gamma = [30, 60]^\circ$.

These graphs additionally illustrate the mixed $(\pi 3s_{1/2}, \pi 2d_{3/2})$ composition of the ground state wavefunction. In Fig. 11(b), the remaining two groups at $\epsilon > 0.2$ lie in regions where $I = \frac{3}{2}^+$, and come close to boundaries with $I = \frac{1}{2}^+$ regions. This shows that while the ground state spin is $I = \frac{1}{2}^+$ due to the dominant $\pi 1s_{1/2}$ component, the magnetic dipole moment is intermediate between values expected for pure $\pi 1s_{1/2}$ and $\pi 2d_{3/2}$ states, due to the $\pi 2d_{3/2}$ admixture in the wavefunction [36].

This region of interest may be further constrained by studying the calculated energy levels. The experimental spectrum summarised in Table 2 indicates a low-lying $I = \frac{3}{2}^+$ state, and small energy gap of 17 keV between a $\frac{9}{2}^+$ state and a $\frac{13}{2}^+$ state. Fig. 12(a) shows the calculated energy levels of these spins. Their variation with γ at fixed $\epsilon = 0.12$ (approximately the centre of the region of interest) is plotted.

The low-lying $\frac{3}{2}^+$ state is approximately reproduced at all γ (see Fig. 12(a)), but the calculated $\frac{13}{2}^+$ energies are overestimated. The discrepancy may be due to the input effective core $E(2^+)$, which assumes that the core moment of inertia will be unaffected by the odd proton. However, if the odd proton increases the moment of inertia, the excitation energies will be scaled down, in accordance with Eq. 13a. Decreasing $E(2^+)$ from 0.17 MeV to 0.12 MeV brings the lowest $\frac{13}{2}^+$ state closer to the experimental value (see Fig. 12(b)) however the $\frac{9}{2}^+$ states are also lowered.

For this reason, the energy gap between the lowest $\frac{13}{2}^+$ and third $\frac{9}{2}^+$ states is used to further constrain the region of interest. This quantity is shown in Fig. 12(c), and the small gap is approximately reproduced in the range $\gamma = [20, 40]^\circ$. In combination with the ground state spin and magnetic dipole moment data, this limits the predicted deformation of $^{179}\text{Au}_{100}$ to the range $\epsilon = [0.06, 0.18]$ and $\gamma = [30, 40]^\circ$. Table 2 compares

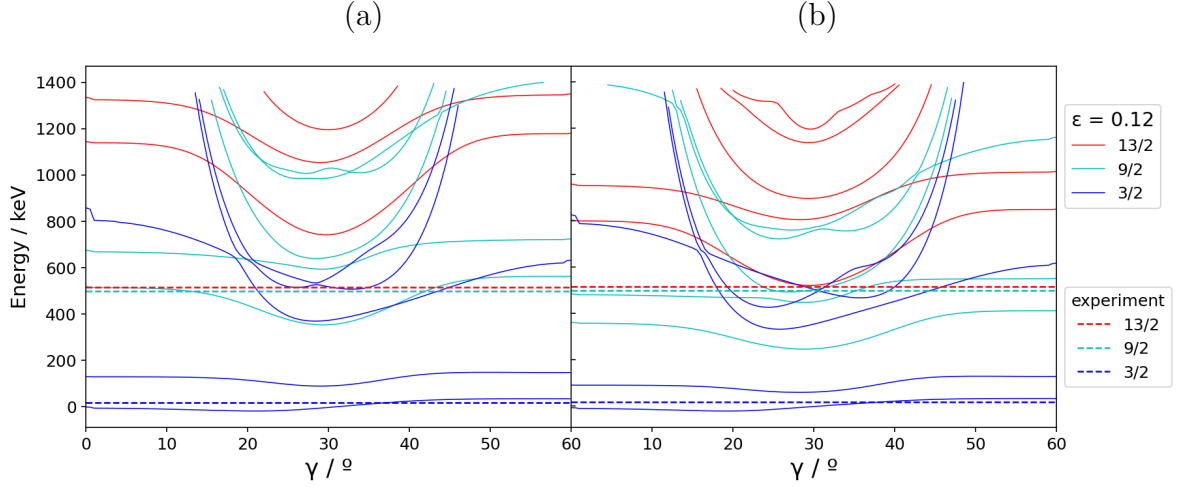


Figure 12: Variation in the calculated $I = \frac{3}{2}^+, \frac{9}{2}^+, \frac{13}{2}^+$ energy levels using (a) effective core $E(2^+) = 0.17$ MeV, and (b) $E(2^+) = 0.14$ MeV. The triaxiality (γ) is varied, with fixed $\varepsilon = 0.12$. The dashed lines indicates the experimental values. All energies are relative to the lowest $\frac{1}{2}^+$ state (the expected ground state). The lowest $\frac{3}{2}^+$ state is close to the experimental value in both (a) and (b). In (b), the energy levels are depressed due to the smaller value of $E(2^+)$. This brings the lowest $\frac{13}{2}^+$ state closer to the experimental value than in (a). The third $\frac{9}{2}^+$ state is then close to the experimental value at $\gamma \approx 30^\circ$ in (b).

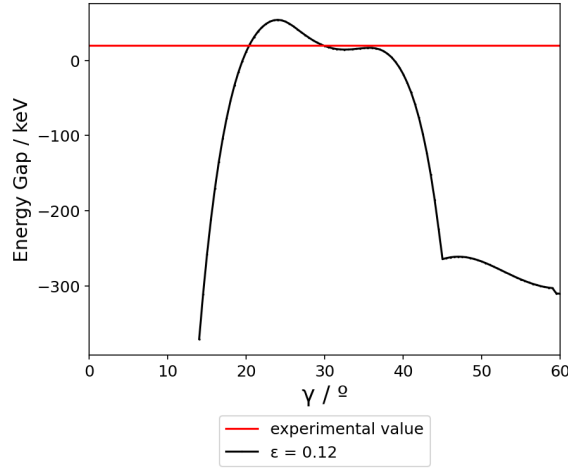


Figure 13: Variation in the calculated energy gap between the lowest $\frac{13}{2}^+$ and third $\frac{9}{2}^+$ energy levels (as seen in Fig. 12(b)). The triaxiality (γ) is varied, with fixed $\varepsilon = 0.12$, and $E(2^+) = 0.14$ MeV. The calculated value is close to the experimental value (the red line) for $\gamma \approx [20, 40]^\circ$.

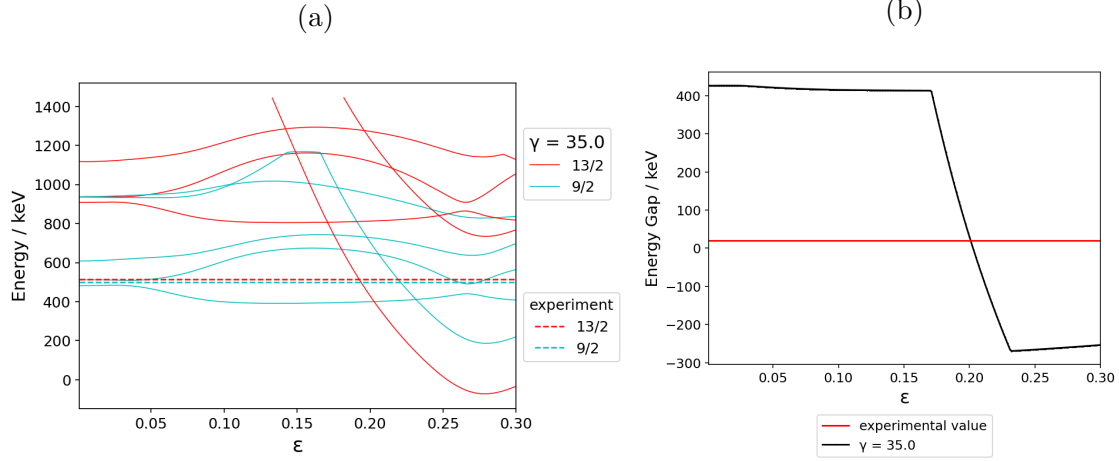


Figure 14: Variation of (a) calculated $I = \frac{3}{2}^+$, $\frac{9}{2}^+$, $\frac{13}{2}^+$ energy levels. ε is varied, with fixed $\gamma = 35^\circ$, and $E(2^+) = 0.17$ MeV. All energies are relative to the lowest $\frac{1}{2}^+$ state (the expected ground state). The dashed lines indicate the experimental values. In (a), an intruder $\frac{13}{2}^+$ state crosses the experimental value at $\varepsilon \approx 0.19$. At this deformation the lowest $\frac{9}{2}^+$ state is close to the experimental value. In (b), the calculated energy gap between the lowest $\frac{13}{2}^+$ and lowest $\frac{9}{2}^+$ energy levels is shown. This crosses the experimental value (the red line) at $\varepsilon \approx 0.20$.

the experimental values with the calculated values at $(\varepsilon, \gamma) = (0.12, 35^\circ)$. Uncertainties are estimated from the range of values calculated across this region.

In this interpretation, the first and second $\frac{9}{2}^+$ states may be states which have not yet been observed experimentally. These two states have dominant spin projections of $K = \frac{3}{2}^+$ and $\frac{5}{2}^+$ respectively. Meanwhile, the third $\frac{9}{2}^+$ state and lowest $\frac{13}{2}^+$ states have dominant spin projections of $K = \frac{7}{2}^+$ and $\frac{3}{2}^+$. All four of these states have the same dominant single particle state $|\nu\rangle$ comprised of three Nilsson states, $\frac{1}{2}[411]$, $\frac{1}{2}[400]$, and $\frac{3}{2}[402]$. We note that the different dominant spin projections (K) of the third $\frac{9}{2}^+$ and lowest $\frac{13}{2}^+$ states may be an indication that they do not belong to the same rotational band, however the classification of states into rotational bands is somewhat ambiguous at high triaxiality due to strong mixing of K -states.

However, other interpretations are possible. A different deformation may better reproduce the experimental excitation energies without the need to adjust the core moment of inertia or disregard the first two calculated $\frac{9}{2}^+$ states.

Plotting variation in ε with fixed $\gamma = 35^\circ$ reveals an intruder $\frac{13}{2}^+$ state which lies between the first and second $\frac{9}{2}^+$ states in the range $\varepsilon = [0.18, 0.20]$ (see Fig. 14(a)). This is a nearly pure Nilsson state $\frac{13}{2}[606]$, originating from the spherical $\pi 1i_{13/2}$ orbital above the $Z = 82$ proton shell closure. In this region the energy gap to the *lowest* $\frac{9}{2}^+$ state becomes small (see Fig. 14(b)). The constraint of the ground state magnetic dipole moment combined with the steep slope of the intruder state limits the region of interest to $\varepsilon = [0.18, 0.20]$, $\gamma = [30, 40]^\circ$.

On the right of Table 2, the calculated values at $(\varepsilon, \gamma) = (0.19, 35^\circ)$ are compared with experimental values. The calculated magnetic dipole moment of the ground state

is underestimated here, however this could be attributed to an unrepresentative choice of the effective g -factor. Overall, this interpretation reduces the fractional errors in the calculated values, and furthermore does not require the assumption that the experimental data is incomplete. However, it does not appear to belong to the same rotational band as any of the low-lying $\frac{9}{2}^+$ states, on account of its very different dominant single particle state. Therefore, we propose that the initial estimate of $(\varepsilon, \gamma) = (0.12, 35)$ is the better description of this nucleus in the PTRM.

Both interpretations are approximately supported by the results of the PES calculations $(\varepsilon, \gamma) = (0.150, 30^\circ)$ in [38], and the estimations $(\varepsilon, \gamma) = (0.18, 25\text{--}30^\circ)$ of [36]. All agree that a high degree of triaxiality in $^{179}_{79}\text{Au}_{100}$ is likely.

4.3 ^{177}Pt

$^{177}_{78}\text{Pt}_{99}$ is neutron deficient, with a substantial number of both valence protons and neutrons. We therefore expect to see deformation in the ground state. An experimental measurement of its magnetic dipole moment has not yet been made, however its excitation spectrum has been measured [40]. The ground state ($I^\pi = \frac{5}{2}^-$) is the band head of the rotational spectrum summarised on the left of Table 3 for spins up to $I = \frac{13}{2}$.

An effective core $E(2^+) = 0.17$ MeV was used, corresponding to the $^{178}_{78}\text{Pt}_{100}$ core [39]. This choice was found to provide a better agreement than the higher value from the other neighbour, $^{176}_{78}\text{Pt}_{98}$. The calculated ground state spins for a mesh of 820 data points up to $\varepsilon = 0.50$ are shown in Fig. 15(a). Two regions of interest can be seen, firstly in the range $\varepsilon \approx [0.2, 0.35]$, $\gamma \approx [15, 35]^\circ$, and secondly a larger area spanning $\varepsilon \approx [0.25, 0.45]$, $\gamma \approx [35, 60]^\circ$.

Energy levels were calculated over several cross sections in the deformation plane. A consistent pattern emerged in all cases, where the lowest energy state of each spin provided the best agreement to experimental data. A typical example is shown in Fig. 15(b), which shows the variation in the calculated energies with γ at fixed $\varepsilon = 0.30$.

Table 3: Comparison of the experimental properties of $^{177}_{78}\text{Pt}_{99}$ with the calculated values at $(\varepsilon, \gamma) = (0.28, 27^\circ)$. The low energy states of the ground rotational band are listed in this table; the full spectrum can be found in [40]. Uncertainties are quoted in brackets (see main text).

Low-lying negative-parity states	Energy relative to the $5/2^-$ ground state / keV	
	Experimental value	Calculated at $(0.28, 27^\circ)$
$13/2^-$	491.8	520(20)
$11/2^-$	336.3	350(20)
$9/2^-$	197.4	180(10)
$7/2^-$	81.2	70(10)
$5/2^-$	0	0

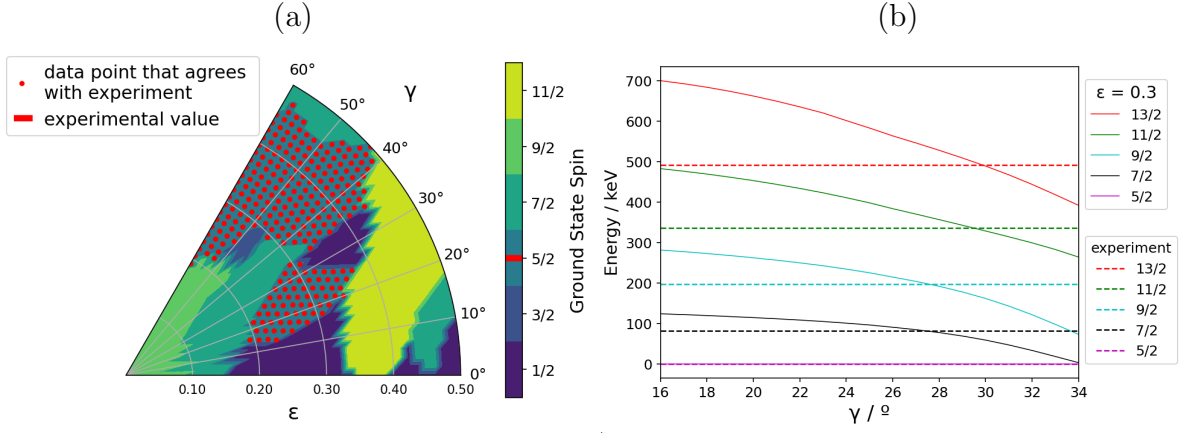


Figure 15: Calculated (a) ground state spins, and (b) excitation energies of $^{177}_{78}\text{Pt}_{99}$. Data points which agree with the experimental ground state spin ($I = \frac{5}{2}$) are marked in red on (a). The dashed lines in (b) show the experimental values of the excitation energies. There are two areas in (a) where the ground state spin is reproduced. (b) is an example of the energy levels at $\varepsilon = 0.3$, $\gamma = [16, 34]^\circ$, and a similar pattern is observed at other deformations in the regions of correct ground state spin. Only the lowest energy state of each spin is shown. The $\frac{7}{2}^+$ and $\frac{9}{2}^+$ states cross their experimental values at $\gamma \approx 27^\circ$. The calculated $\frac{11}{2}^+$ and $\frac{13}{2}^+$ energies are overestimated at that deformations; this discrepancy can be attributed to the rigid rotor assumption.

In this figure we can see the effects of the rigid rotor approximation (discussed in Sec. 2.2), as a triaxiality of $\gamma \approx 27.5^\circ$ is a good agreement for the lower spin states ($I = \frac{7}{2}, \frac{9}{2}$), whereas the energies of the higher spin states ($I = \frac{11}{2}, \frac{13}{2}$) are increasingly overestimated at this deformation. The agreement of these results could be improved with a variable moment of inertia (VMI) approach [15], which would model the increase in moment of inertia with rotation, caused by the short range of the nuclear force [9].

A similar pattern was observed in several other locations across the regions of interest. In all cases, the lowest energy states of each spin have the expected wavefunction composition. As members of a single rotational band, these levels all have the same dominant $K = \frac{5}{2}$ component. Furthermore, they are all built predominantly upon the same single particle state (Eq. 11) which is dominated by Nilsson states $\frac{5}{2}[523]$, $\frac{5}{2}[512]$, and $\frac{5}{2}[532]$. Previous analysis [40] has assigned $\frac{5}{2}[512]$ as the dominant component of the ground state wavefunction, based on experimental g -factor values and enhanced transition strengths to daughter nuclei of that configuration.

In order to identify the best fitting deformation, the root-mean-square (RMS) error in the calculated energies was determined at each deformation. The results are plotted on Fig. 16, indicating that a deformation of $\varepsilon = [0.28 \pm 0.04]$ and $\gamma = [28 \pm 4]^\circ$ minimises the RMS error. The position of this minimum is stable with respect to the effective core moment of inertia for $E(2^+) = [0.14, 0.30]$ MeV.

The results of the calculations at $(\varepsilon, \gamma) = (0.28, 27^\circ)$ are summarised on the right of Table 3, with uncertainties estimated from the range in this region. A scaling factor of $\chi = 0.7$ is used to calculate the magnetic dipole moment and spectroscopic electric quadrupole moment as $\mu_D = [-0.38 \pm 0.04] \mu_N$ and $Q_s = [2.7 \pm 0.2] eb$ respectively. Performing the first experimental measurements of these quantities would assist with

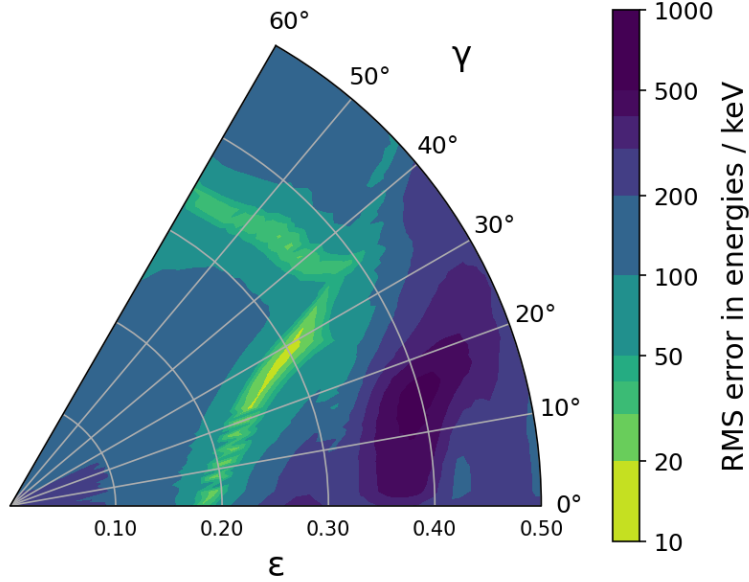


Figure 16: RMS error in excitation energies of $^{177}_{78}\text{Pt}_{99}$ across the deformation plane. The colour map uses a log scale, highlighting the best fitting values (with RMS error less than 20 keV) in yellow at $(\varepsilon, \gamma) \approx (0.28, 27^\circ)$.

constraining the deformation parameters of this nucleus.

Previous analysis [40] using an axially symmetric particle-rotor model (among other methods) predicted a quadrupole deformation of $\varepsilon = 0.21$. In Fig. 16, if we restrict our view to the $\gamma = 0^\circ$ edge of the sector, we can see that the best agreement to experimental results is obtained at $\varepsilon \approx 0.2$, in agreement with that paper. The authors applied a Coriolis attenuation scaling factor of 0.65 to reproduce the experimental ground state spin. However, the deformation deduced in the present work (with the inclusion of the triaxiality degree of freedom) reproduces the experimental ground state spin without a need to artificially reduce Coriolis mixing.

5 Conclusions

In summary, the Particle-plus-Triaxial-Rotor Model has been applied to study three nuclei in the lead region. The spectroscopic properties of a nucleus are calculated, assuming a range of quadrupole and triaxial deformations. Comparisons of the calculated ground state spin and magnetic dipole moment with experimental values (where available) are used to identify regions of interest. Energy levels are then calculated over cross sections in these regions to further constrain the possible deformations. These can be analysed in terms of shell model single particle excitations, rotational bands (including energy level ordering and spacing), and wavefunction compositions.

The PTRM has been shown to be capable of describing both the low-lying single particle excitations in the near-spherical nucleus $^{207}_{82}\text{Pb}_{125}$, as well as the ground rotational band of the well-deformed nucleus $^{177}_{78}\text{Pt}_{99}$. The latter was deduced to have a deformation of $\varepsilon = [0.28 \pm 0.04]$, $\gamma = [27 \pm 3]^\circ$, making use of the triaxial degree of freedom to improve upon previous estimates which were limited to axial symmetry. This conclusion

should be tested by future measurements of the magnetic dipole and electric quadrupole moments of the ground state, which are predicted to be $\mu_D = [-0.38 \pm 0.04] \mu_N$ and $Q_s = [2.7 \pm 0.2] eb$ respectively in these calculations.

The results for $^{179}_{79}\text{Au}_{100}$ were less conclusive, with two possibilities emerging. A deformation of $\varepsilon = [0.12 \pm 0.06]$, $\gamma = [35 \pm 5]^\circ$ would imply that there exist two $\frac{9}{2}^+$ states at energies below 498 keV which have yet to be measured experimentally. On the other hand, if we assume a deformation of $\varepsilon = [0.19 \pm 0.01]$, $\gamma = [35 \pm 5]^\circ$ then we would expect the experimental $\frac{9}{2}^+$ state to have dominant configuration $\frac{13}{2}[606]$. A more in-depth study of this state is required to confirm the composition of its wavefunction.

There are several key limitations to the calculations. The PTRM is not capable of modelling odd-multipole deformations (most commonly octupoles). Therefore, care must be taken when analysing nuclei in regions with known or suspected octupole deformations. There are two additional degrees of freedom in the inputs which form the primary source of uncertainty in the results. The effective core moment of inertia is typically assumed to be the same as that of the even-even core, however the validity of this choice is not clear, and further work is required to investigate the effect of the odd particle on the core rotational spectrum. Secondly, the use of a scaling factor on the free intrinsic spin g -factor of the odd particle to approximately model complex behaviour is a weakness of the PTRM, and it could be improved with a less schematic treatment.

The analysis performed in this work is heavily reliant on experimental data for the nuclei studied. Measurements of the ground state spin, parity, and the low-lying states of at least one rotational band are essential, and measurements of the ground state magnetic dipole moment and electric quadrupole moments are highly beneficial for constraining the results.

With this method, we have demonstrated that the *combined* picture of these commonly measured properties can serve as a signature of triaxiality, in nuclei where the triaxial degree of freedom is necessary to reproduce this full picture. A single property alone may not be sufficient to draw conclusions, however by considering the nucleus as a whole, we can see the role of triaxiality in generating the properties we observe. The approach taken in this work may be applied to many more isotopes across the nuclear chart, for a broader picture of how nuclear deformation depends on particle number. Such data may motivate and support future theoretical models, contributing to our understanding of proton-neutron interactions and nuclear deformations.

For this purpose, the Python interface to the PTRM codes developed in this work may assist future users of this model, by facilitating a more efficient workflow. On an 8-core computer, 820 deformations can be made within 15 seconds, and this can be scaled to thousands of calculations in a similar time frame on a larger machine. There are several ways in which the script could be extended for future applications. Applying the variable moment of inertia (VMI) settings would ease the rigid rotor approximation, and improve the agreement of calculated energy levels to their experimental values. Classification of calculated energy levels into individual rotational bands based on the calculated quantum numbers would assist with the assignment of experimental values to calculated states, as would labelling of the calculated states with the dominant components of the wavefunctions. Additionally, analysis of the reported electromagnetic transition strengths and rates could provide insight into the influence of triaxiality on these decay properties. Future work in this direction holds promise for an improved understanding of the triaxial degree of freedom in nuclei.

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References

- [1] Nguyen Le Anh et al. “Landscape of nuclear deformation softness with spherical quasiparticle random-phase approximation”. In: *Physical Review C* 109.2 (Feb. 2024). ISSN: 2469-9993. DOI: 10.1103/physrevc.109.024313. URL: <http://dx.doi.org/10.1103/PhysRevC.109.024313>.
- [2] V.G. Soloviev, Ch. Stoyanov, and V.V. Voronov. “Nuclear properties in the lead region within the quasiparticle-phonon nuclear model”. In: *Nuclear Physics A* 399.1 (May 1983), pp. 141–162. ISSN: 0375-9474. DOI: 10.1016/0375-9474(83)90599-7. URL: [http://dx.doi.org/10.1016/0375-9474\(83\)90599-7](http://dx.doi.org/10.1016/0375-9474(83)90599-7).
- [3] R C Thompson et al. “High-resolution measurements of isotope shifts and hyperfine structure in stable and radioactive lead isotopes”. In: *Journal of Physics G: Nuclear Physics* 9.4 (Apr. 1983), pp. 443–458. ISSN: 0305-4616. DOI: 10.1088/0305-4616/9/4/010. URL: <http://dx.doi.org/10.1088/0305-4616/9/4/010>.
- [4] Paresh Mukherjee and Bernard L. Cohen. “Nuclear Structure Studies in the Lead Region with Stripping Reactions”. In: *Physical Review* 127.4 (Aug. 1962), pp. 1284–1296. ISSN: 0031-899X. DOI: 10.1103/physrev.127.1284. URL: <http://dx.doi.org/10.1103/PhysRev.127.1284>.
- [5] A. N. Andreyev et al. “A triplet of differently shaped spin-zero states in the atomic nucleus 186Pb”. In: *Nature* 405.6785 (May 2000), pp. 430–433. ISSN: 1476-4687. DOI: 10.1038/35013012. URL: <http://dx.doi.org/10.1038/35013012>.
- [6] Paul E. Garrett, Magda Zielińska, and Emmanuel Clément. “An experimental view on shape coexistence in nuclei”. In: *Progress in Particle and Nuclear Physics* 124 (May 2022), p. 103931. ISSN: 0146-6410. DOI: 10.1016/j.ppnp.2021.103931. URL: <http://dx.doi.org/10.1016/j.ppnp.2021.103931>.
- [7] S. Frauendorf and A.O. Macchiavelli. “Overview of neutron–proton pairing”. In: *Progress in Particle and Nuclear Physics* 78 (Sept. 2014), pp. 24–90. ISSN: 0146-6410. DOI: 10.1016/j.ppnp.2014.07.001. URL: <http://dx.doi.org/10.1016/j.ppnp.2014.07.001>.
- [8] S. G. Nilsson. “Binding States of Individual Nucleons in Strongly Deformed Nuclei”. In: *Matematisk-Fysiske Meddelelser* 29.16 (1955).
- [9] Ingemar Ragnarsson and Sven Gösta Nilsson. *Shapes and Shells in Nuclear Structure*. Cambridge University Press, June 1995. ISBN: 9780511563973. DOI: 10.1017/cbo9780511563973. URL: <http://dx.doi.org/10.1017/CB09780511563973>.
- [10] J. Timár et al. “Triaxiality-related nuclear phenomena in the A = 100 mass region”. In: *Journal of Physics: Conference Series* 1555.1 (May 2020), p. 012025. ISSN: 1742-6596. DOI: 10.1088/1742-6596/1555/1/012025. URL: <http://dx.doi.org/10.1088/1742-6596/1555/1/012025>.

- [11] Costel Marian Petrache. *Chirality and Wobbling in Atomic Nuclei*. CRC Press, Nov. 2024. ISBN: 9781032691633. DOI: 10.1201/9781032691633. URL: <http://dx.doi.org/10.1201/9781032691633>.
- [12] N.V. Zamfir and R.F. Casten. “Signatures of gamma-softness or triaxiality in low energy nuclear spectra”. In: *Physics Letters B* 260.3–4 (May 1991), pp. 265–270. ISSN: 0370-2693. DOI: 10.1016/0370-2693(91)91610-8. URL: [http://dx.doi.org/10.1016/0370-2693\(91\)91610-8](http://dx.doi.org/10.1016/0370-2693(91)91610-8).
- [13] Nazira Nazir et al. “Microscopic aspects of gamma-softness in atomic nuclei”. In: *Physical Review C* 107.2 (Feb. 2023). ISSN: 2469-9993. DOI: 10.1103/physrevc.107.1021303. URL: <http://dx.doi.org/10.1103/PhysRevC.107.L021303>.
- [14] S.E. Larsson, G. Leander, and I. Ragnarsson. “Nuclear core-quasiparticle coupling”. In: *Nuclear Physics A* 307.2 (Sept. 1978), pp. 189–223. ISSN: 0375-9474. DOI: 10.1016/0375-9474(78)90613-9. URL: [http://dx.doi.org/10.1016/0375-9474\(78\)90613-9](http://dx.doi.org/10.1016/0375-9474(78)90613-9).
- [15] Ingemar Ragnarsson and Paul B. Semmes. “Description of nuclear moments and nuclear spectra in the particle-rotor model”. In: *Hyperfine Interactions* 43.1–4 (Dec. 1988), pp. 423–440. ISSN: 1572-9540. DOI: 10.1007/bf02398323. URL: <http://dx.doi.org/10.1007/BF02398323>.
- [16] Tomás R. Rodríguez and J. Luis Egido. “Triaxial angular momentum projection and configuration mixing calculations with the Gogny force”. In: *Physical Review C* 81.6 (June 2010). ISSN: 1089-490X. DOI: 10.1103/physrevc.81.064323. URL: <http://dx.doi.org/10.1103/PhysRevC.81.064323>.
- [17] V. Kumar et al. “Spectroscopic study of ^{97}Mo , ^{99}Mo , and ^{101}Mo ”. In: *Physical Review C* 110.4 (Oct. 2024). ISSN: 2469-9993. DOI: 10.1103/physrevc.110.044308. URL: <http://dx.doi.org/10.1103/PhysRevC.110.044308>.
- [18] M. Balogh et al. “New collective structures in ^{179}Au and their implications for the triaxial deformation of the ^{178}Pt core”. In: *Physical Review C* 106.6 (Dec. 2022), p. 064324. ISSN: 2469-9993. DOI: 10.1103/physrevc.106.064324. URL: <http://dx.doi.org/10.1103/PhysRevC.106.064324>.
- [19] S. Chakraborty et al. “Exploring the possibility of wobbling motion in ^{129}Ba ”. In: *Physical Review C* 110.2 (Aug. 2024). ISSN: 2469-9993. DOI: 10.1103/physrevc.110.024324. URL: <http://dx.doi.org/10.1103/PhysRevC.110.024324>.
- [20] T. Krasta, L. Simonova, and D. Riekstina. “Newby Shifts in Odd-Odd Transitional Nuclei at $A = 190$ ”. In: *Bulletin of the Russian Academy of Sciences: Physics* 85.12 (Dec. 2021), pp. 1466–1470. ISSN: 1934-9432. DOI: 10.3103/s1062873821120170. URL: <http://dx.doi.org/10.3103/S1062873821120170>.
- [21] P. Petkov et al. “Particle-triaxial rotor calculations for the low-lying states with negative parity in ^{193}Os ”. In: *The European Physical Journal A* 36.2 (May 2008), pp. 127–130. ISSN: 1434-601X. DOI: 10.1140/epja/i2007-10586-y. URL: <http://dx.doi.org/10.1140/epja/i2007-10586-y>.
- [22] A. K. Mondal et al. “Investigation of the low- and medium-spin level structure in ^{77}As ”. In: *Physical Review C* 107.6 (June 2023). ISSN: 2469-9993. DOI: 10.1103/physrevc.107.064320. URL: <http://dx.doi.org/10.1103/PhysRevC.107.064320>.

- [23] D. Ivanova et al. “Structure of the low-lying states in 99Pd, 101Pd, 103Pd”. In: *Physical Review C* 105.3 (Mar. 2022). ISSN: 2469-9993. DOI: 10.1103/physrevc.105.034337. URL: <http://dx.doi.org/10.1103/PhysRevC.105.034337>.
- [24] S. Aydin et al. “Lifetime measurements and evidence for triaxial nuclear shapes in 127Cs”. In: *Physical Review C* 104.5 (Nov. 2021). ISSN: 2469-9993. DOI: 10.1103/physrevc.104.054309. URL: <http://dx.doi.org/10.1103/PhysRevC.104.054309>.
- [25] Umakant Lamani, Pragya Das, and S. G. Wahid. “Study of signature inversion in 123, 125I using Particle Rotor Model calculations”. In: *International Journal of Modern Physics E* 30.07 (July 2021), p. 2150062. ISSN: 1793-6608. DOI: 10.1142/s0218301321500622. URL: <http://dx.doi.org/10.1142/S0218301321500622>.
- [26] Z. Housni et al. “Low-energy band structures in light odd-A La isotopes using the quasiparticle-phonon coupling plus rotor approach”. In: *Physical Review C* 103.4 (Apr. 2021). ISSN: 2469-9993. DOI: 10.1103/physrevc.103.044310. URL: <http://dx.doi.org/10.1103/PhysRevC.103.044310>.
- [27] Sven Gösta Nilsson et al. “On the nuclear structure and stability of heavy and superheavy elements”. In: *Nuclear Physics A* 131.1 (June 1969), pp. 1–66. ISSN: 0375-9474. DOI: 10.1016/0375-9474(69)90809-4. URL: [http://dx.doi.org/10.1016/0375-9474\(69\)90809-4](http://dx.doi.org/10.1016/0375-9474(69)90809-4).
- [28] K.S. Krane. *Introductory Nuclear Physics*. Wiley John & Sons, New York, 1988. ISBN: 9780471805533.
- [29] P. Möller and J.R. Nix. “Nuclear pairing models”. In: *Nuclear Physics A* 536.1 (Jan. 1992), pp. 20–60. ISSN: 0375-9474. DOI: 10.1016/0375-9474(92)90244-e. URL: [http://dx.doi.org/10.1016/0375-9474\(92\)90244-E](http://dx.doi.org/10.1016/0375-9474(92)90244-E).
- [30] J. Bardeen, L. N. Cooper, and J. R. Schrieffer. “Microscopic Theory of Superconductivity”. In: *Physical Review* 106.1 (Apr. 1957), pp. 162–164. ISSN: 0031-899X. DOI: 10.1103/physrev.106.162. URL: <http://dx.doi.org/10.1103/PhysRev.106.162>.
- [31] I. Ragnarsson P.B. Semmes. *The Particle + Triaxial Rotor Model: A User’s Guide*. Presented at the Nuclear Structure Theory Workshop, Oak Ridge, TN, June 1992.
- [32] B. R. Mottelson A. Bohr. *Nuclear Structure*. W. A. Benjamin, Inc., 1975.
- [33] Verena Fella et al. “Magnetic moment of 207Pb and the hyperfine splitting of 207Pb 81+”. In: *Physical Review Research* 2.1 (Mar. 2020). ISSN: 2643-1564. DOI: 10.1103/physrevresearch.2.013368. URL: <http://dx.doi.org/10.1103/PhysRevResearch.2.013368>.
- [34] M. Kadi et al. “Decay properties of states populated with the 207Pb(n,n’ γ) reaction and weak coupling in 207Pb”. In: *Physical Review C* 61.3 (Feb. 2000). ISSN: 1089-490X. DOI: 10.1103/physrevc.61.034307. URL: <http://dx.doi.org/10.1103/PhysRevC.61.034307>.
- [35] M. Rejmund et al. “Particle octupole-vibration coupling near 208Pb”. In: *The European Physical Journal A* 8.2 (July 2000). ISSN: 1434-601X. DOI: 10.1007/s100500070102. URL: <http://dx.doi.org/10.1007/s100500070102>.

- [36] J.G. Cubiss et al. “Change in structure between the $I=1/2$ states in ^{181}Tl and $^{177}, ^{179}\text{Au}$ ”. In: *Physics Letters B* 786 (Nov. 2018), pp. 355–363. ISSN: 0370-2693. DOI: 10.1016/j.physletb.2018.10.005. URL: <http://dx.doi.org/10.1016/j.physletb.2018.10.005>.
- [37] M. Venhart et al. “De-excitation of the strongly coupled band in ^{177}Au and implications for core intruder configurations in the light Hg isotopes”. In: *Physical Review C* 95.6 (June 2017). ISSN: 2469-9993. DOI: 10.1103/physrevc.95.061302. URL: <http://dx.doi.org/10.1103/PhysRevC.95.061302>.
- [38] P. Möller et al. “Nuclear shape isomers”. In: *Atomic Data and Nuclear Data Tables* 98.2 (Mar. 2012), pp. 149–300. ISSN: 0092-640X. DOI: 10.1016/j.adt.2010.09.002. URL: <http://dx.doi.org/10.1016/j.adt.2010.09.002>.
- [39] F. G. Kondev et al. “High-spin collective structures in ^{178}Pt ”. In: *Physical Review C* 61.4 (Mar. 2000). ISSN: 1089-490X. DOI: 10.1103/physrevc.61.044323. URL: <http://dx.doi.org/10.1103/PhysRevC.61.044323>.
- [40] G.D. Dracoulis et al. “Intrinsic states and rotational bands in ^{177}Pt ”. In: *Nuclear Physics A* 510.3 (Apr. 1990), pp. 533–556. ISSN: 0375-9474. DOI: 10.1016/0375-9474(90)90065-t. URL: [http://dx.doi.org/10.1016/0375-9474\(90\)90065-t](http://dx.doi.org/10.1016/0375-9474(90)90065-t).